

Matrix Compositions: Exact Spectra, Hankel Products, and Uniform Asymptotics

OEIS A261781, its empty-row transition,
and refined asymptotics for A261784

Part II: finite spectra, uniform asymptotics, and two Poisson limits

Part I: OpenAI ChatGPT

Research prepared for Vladimir Reshetnikov

Part II: a research report prepared for Vladimir Reshetnikov

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Abstract

Let $T(n, k)$ count nonnegative integer matrices of total sum n , with k ordered rows, an arbitrary number of ordered columns, and no zero row or column. These are the entries of OEIS A261781. We establish a collection of exact and asymptotic results for this family. First, its minimal eventual constant-coefficient recurrence has order $k(k+1)/2$, and its full-rank shifted Hankel determinants admit an explicit positive product formula, including a polynomial refinement marking the number of columns. Second, we give an exponential relative error bound for the unrestricted count that is uniform in the number of rows. This yields a Poisson empty-row law at $n = k(\log k + s)$, an explicit first correction with a summable error of order $(\log k)^2/k^2$, and the sharp total-variation rate with its leading constant. We also prove joint Gaussian–Poisson independence for column count and empty-row count in this window. Third, for n/k in compact subsets of $(1, \infty)$, a positive Stirling-number transform gives a uniform saddle expansion to every fixed order. For A261784, this identifies the exact amplitude in the recorded leading equivalent and gives the first relative correction. Finally, fixing the number of columns produces a different explicit coverage threshold.

Attribution matters here: the generating functions, a recurrence of the stated order, and fixed-row asymptotics were already published by Munarini, Poneti, and Rinaldi. Thus the two conjecture labels still present in A261781 lead in part to a reconciliation with prior literature. The main extensions of this article are the factored determinant and the uniform distributional and asymptotic results. Full proofs, reproducible computations, a bounded source audit, and further research questions are included. These are unrefereed mathematical proofs; no proof-assistant verification or historical-priority guarantee is claimed.

Editorial note (3 October 2026, batch 85B). This report is built from two manuscripts of ProveIt’s incoming-reports intake, written independently of each other on the same subject: manuscript 06 of batch 85 (Part I, the base, whose abstract stands above) and manuscript 01 of batch 85 (Part II). It is AI-assisted, unrefereed and not formalized. Where the two manuscripts prove the same statement, it is printed once, in Part I, in manuscript 06’s form, with a bracketed note marked “[Added 3 October 2026, batch 85B]” that names manuscript 01’s version and its route; Part II (Section 11 onwards) prints everything that manuscript 01 adds. *All of Part II assumes*

the column weight $u = 1$: manuscript 01 never weights columns. The two manuscripts permute their Greek letters; everything here is printed in manuscript 06's letters, and the dictionary of Section 1.5 discloses every rename. Manuscript 01 presented its recurrence theorem as a proof of the OEIS order conjecture; Munarini, Poneti and Rinaldi had already proved a recurrence of that order with the same denominator [4, Proposition 25 and Remark 26, p. 16], so only its minimality is new (Section 11.2). Part I's labels carry the prefix `mxc:`, Part II's the prefix `mxc:tp:`; Appendix A records the provenance and every editorial change.

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Part I

Exact spectra, Hankel products, and uniform asymptotics

1 The problem, the sources, and the contribution

1.1 Why this family is worth studying

One generating function can behave quite differently when a parameter is fixed or grows with the coefficient index. Matrix compositions provide an especially explicit setting in which to study this difference. For fixed k , the sequence is a finite sum of exponential terms and has an exact linear recurrence. At $n \asymp k$, requiring every row to be used is a rare event, and a saddle point governs its count. At $n = k(\log k + O(1))$, the same requirement has a nondegenerate limiting probability. These regimes cannot be interchanged by simply substituting a growing k into a fixed- k asymptotic.

The current A261781 entry attributes to Václav Kotěšovec two conjectures dated October 14, 2017: a column recurrence order $k(k+1)/2$, and the limit $d(k+1) - d(k) \rightarrow 1/\log 2$ for its exponential growth constants [2]. Its companion A261780 counts the same matrices while allowing zero rows [1]. The diagonal $T(2N, N)$ is A261784 [3].

1.2 What was already known

The primary enumerative source is *Matrix Compositions* by Munarini, Poneti, and Rinaldi [4], preceded by a 2006 conference paper [5]. In the 2009 article, Proposition 1 gives the generating function, Propositions 12–13 give a root-of-unity formula and fixed-row asymptotics, Propositions 22–24 give the no-zero-row inclusion–exclusion formula, Proposition 25 gives a recurrence of order $\binom{k+1}{2}$, and Proposition 29 gives the positive Stirling/Fubini transform used below. Our elementary derivations of these identities make the article self-contained; the identities themselves are attributed to those authors.

Louchard [6] studies fixed-row probabilistic asymptotics, including the number of columns and column-size statistics. An empty *column size* in that work is distinct from a zero *row* here. The A261784 entry already records a leading equivalent with numerical amplitude and an explicit exponential constant, credited to Kotěšovec in 2017 and updated in 2024 [3].

Accordingly, this article does not claim the discovery of the fixed-row growth formula, recurrence existence, or the leading A261784 exponential rate. The first OEIS statement receives an explicit minimality proof and a careful indexing convention; the second follows from an already known closed form. The stronger developments concern exact determinants and uniform estimates as the number of rows grows.

1.3 Relation to ProveIt and the audit boundary

The target was selected after inspecting the OEIS research in ProveIt [7]. Searches for A261780, A261781, and A261784 found no indexed matching report. The relevant repository hit for matrix compositions was a citation in the report on A260700 and parabolic double cosets, which studies a different enumeration. The repository search exposed indexed commit `ce37e13f4aa16819c87c3ccc611362d15758f2ac`; the search result is not an exhaustive statement about every branch or historical archive. No unformalized repository theorem is assumed in our proofs.

Topic	Existing foundation	Result established here
Recurrences	Order- $\binom{k+1}{2}$ recurrence	Minimal eventual order; complete spectrum; initial impulse
Hankel determinants	Size- k identity for unrestricted matrices	Factored size- $\binom{k+1}{2}$ determinant with column weights
Fixed-row asymptotics	Dominant pole and amplitude	Explicit error uniform in all $k \geq 1$
Empty rows	Exact inclusion–exclusion	Poisson window, first correction, sharp total variation
Column count	Fixed-row Gaussian law	Joint Gaussian–Poisson limit; conditional limits
Diagonal A261784	Recorded leading equivalent	Exact amplitude, first correction, uniform all-orders ray expansion
Fixed column count	Stars-and-bars enumeration	Explicit shifted coverage threshold when columns grow slowly

Table 1: Attribution and the scope of the extensions. “Established here” describes the proofs in this article, not a claim of priority over all literature.

The source audit compared current OEIS records with the primary 2006, 2008, and 2009 works, and checked targeted searches for growing-row, empty-row, and coupon-collection results. It found no matching statement of the specific uniform expansions below. This is a bounded audit, so the responsible claim is a set of fully specified results and proofs, rather than a declaration of a historical breakthrough.

[Added 3 October 2026, batch 85B: the search statement describes the pinned index `ce37e13f4`. At that commit manuscript 01, the source of Part II, had not yet arrived; it now forms Part II of this report, which is filed at `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a261781-matrix-compositions`. Three repository neighbours bear on the method, and none is a dependency. (i) The report on A260700 [10] uses the same two devices as Section 7, a uniform extraction of the Fubini pole at L (its Section `pd:sec:poles`) followed by an outer Stirling transform (its Section `pd:sec:outer`), for a different sequence; it cites [4] as related work only. (ii) The canonical transseries volume proves the exact pole-lattice formula for the Fubini numbers, its Theorem `q2:thm:fubini` [11], of which the estimate (7.5) is the one-pole truncation. (iii) The only ingredient of this report that is formalized anywhere in the repository is the Fubini exponential generating function used in Section 7: the Lean declarations `Fabius.fubini` (the definition $F_q = \sum_b b! \{b^q\}$), `Fabius.egfA_fubini` and `Fabius.two_sub_exp_mul_egfA_fubini`, the identity $(2 - e^t) \sum_q F_q t^q / q! = 1$, in `Analysis/FabiusFunction/Lean/FabiusFunction/OrderedBell.lean` [12]. Nothing else stated in either Part is formalized, and placement beside the repository’s formal projects confers no formal status.]

1.4 Notation and conventions

Throughout, n is total matrix mass, k is the number of rows, and C is the number of columns. Unless stated otherwise, M denotes the number of zero rows in an unrestricted uniformly chosen matrix. Write

$$L = \log 2, \quad a_k = 2^{-1/k}, \quad r_k = 1 - a_k, \quad d_k = r_k^{-1}, \quad c_k = \frac{a_k}{2kr_k}.$$

The empty-row window uses $h = n/k$ and $\eta = ke^{-h}$. The dense-ray regime uses $\lambda = n/k$ and a saddle $r > 0$; these are separate local notations. The falling factorial is $(x)_j = x(x-1) \cdots (x-j+1)$.

All logarithms are natural. Constants in uniform big- O notation may depend on the fixed compact sets specified in each theorem.

We include one empty matrix at $n = 0$ for each fixed height when zero rows are allowed. Thus $A(0, k) = 1$, $A(n, 0) = 0$ for $n > 0$, $T(0, 0) = 1$, and $T(0, k) = 0$ for $k > 0$. These conventions matter for recurrences and Hankel rank at the initial index.

1.5 Notation dictionary for the two manuscripts

[Added 3 October 2026, batch 85B.] Manuscripts 06 (Part I) and 01 (Part II) use the same objects but different letters, and three Greek letters are cyclically permuted between them: manuscript 06's η , λ , ρ are manuscript 01's λ , ρ , η . Copying a formula from one manuscript into the other without translation therefore gives a formula that is wrong, not merely differently named. This report prints *everything* in manuscript 06's letters. Table 2 lists every symbol of manuscript 01 that was renamed for Part II and Part I's notes, with the tempting false reading where one exists; manuscript 01's other symbols are unchanged. Normalizations are unchanged throughout: every rename is a change of letter only. All of Part II is at column weight $u = 1$, so its A , T , c_k , d_k are the unweighted ones of Part I.

Table 2: Manuscript 01's symbols and the letters used for them in this report. “False reading” gives what the manuscript-01 letter means in this report.

Notion	This report (ms. 06)	Ms. 01	False reading if copied
Matrices allowing zero rows	$A(n, k)$	$U(n, k)$	—
Its generating function	$F_k(z; 1)$	$U_k(z)$	—
Zero rows of an unrestricted matrix	M	$Z_{n,k}$	—
Poisson mean $ke^{-n/k}$	η	λ	λ is n/k on dense rays
Ratio n/k on compact rays	λ	ρ	ρ is the pole-bound ratio
Pole-bound ratio $L/\sqrt{L^2 + 8}$	ρ	η	η is the Poisson mean
Dense-ray saddle, $r/(1 - e^{-r}) = \lambda$	r	τ	τ is a formal variable in (7.16); ms. 01's r is $n - k$ or a moment index
Half-saddle mass $Lr/2$	v	μ	$\mu_k(u)$ is an amplitude in Theorem 4.1 and a column mean in (6.1)
Fixed- k constants	$c_k, d_k; d(x)$ at real x	$c(k), d(k),$ $d(x)$	—
Diagonal index	$N,$ $a_N = T(2N, N)$	$m, a(m)$	m indexes d_m and counts columns in Section 8
Diagonal constants	c_*, d_*	c, d	c_k, d_k are the fixed- k constants
Diagonal corrections	$\beta_1 = \beta, \beta_2, \beta_3$	$\delta_1, \delta_2, \delta_3$	ms. 01's “ β_1 ” is $\tilde{\beta}_1 = \beta_1 + \frac{1}{6}$, a different constant
Saddle correction coefficients	$E_j(\lambda)$	$A_m(\rho)$	$A(n, k)$ is a count
Leading saddle approximation	$B(n, k)$	$B(n, k)$	$B, B(u)$ are pole-bound constants
Truncation order of an expansion	J	M	M is the zero-row count
Coverage correction polynomials	$\mathcal{Q}_j(h, \eta; L)$	$Q_j(h, \lambda; L)$	$Q_q(d), Q_{k,u}(z), Q$ are other objects
Coverage exponent	$\Psi(j, h, \epsilon),$ $\epsilon = 1/k$	$G(r, h, \epsilon)$	—
Moment index; point mass	$j; m$	$r; t$	r is the saddle

Notion	This report (ms. 06)	Ms. 01	False reading if copied
Reduced denominator; its degree	$Q_{k,1}(z); D$	$D_k(z); N_k$	D is also $n - Q$ in Section 7.6
Denominator coefficients	q_i	d_r	d_k is a growth constant
Roots and roots of unity	$a_j = 2^{-1/j}, z_{j,\ell}, \omega_j$	$q_j, r_{j,h}, \zeta_j$	q_i are denominator coefficients
Fubini numbers	F_q	F_j	$F_k(z; u)$ is a generating function
Entire Stirling kernel; its coefficients	$W_n(w); \chi_{n,\ell}$	$W_n(v); w_{n,\ell}$	W_N is a diagonal diagnostic, $w_{j,\lambda}$ a Hankel weight
Kernel variable; Euler operator	$w = Lz; D_w$	$v; D_v$	$v = Lr/2$
Kernel correction polynomials	$P_r(w)$	$p_r(v)$	$P_j(t)$ in Section 3 is another polynomial
Kernel-proof variable $1/n$	ϵ	h	$h = n/k$
Kernel-proof auxiliaries	$\Sigma_j(n), \mathbf{a}_j, \mathbf{e}_\ell, \mathbf{r}_\ell, \mathbf{b}_r, \Pi_r(\ell), \mathbf{B}_a, \delta_0$	$S_j(n), A_j, E_\ell, R_\ell, B_r, P_r(\ell), B_a, \delta$	$E_j(\lambda), R_p(w), B, \delta$ of Lemma 7.2
Gaussian functional	$\mathcal{G}_b$	$\mathfrak{G}_b$	—
Cumulant recursion	$\mathcal{K}_j(z, a)$	$K_j(z, a)$	K is a compact set of rays
Stirling index; Stirling defect	$Q_{n,k}; \Delta_{n,k} = n - Q_{n,k}$	$J_{n,k}; \Delta_{n,k}$	J_q is a block count; $\Delta_{n,k}$ is Section 7.6's D
Defect generating variable	ξ	u	u is the column weight
Smooth comparator; its root	$\mathcal{A}^{[J]}(x); x_J(Y)$	$A^{[M]}(x); x_M(Y)$	—
Inverse core	$\Lambda, \varpi = W_0(\alpha\Lambda)$	X, w	X is a Poisson or normalized variable, w a PGF or kernel variable
Integer threshold; its bracket	$\mathcal{N}(Y); \epsilon_J, C_J$	$N(Y); \epsilon_M, C_M$	N is the diagonal index
$\log 2$	L	$L, \text{macro} \backslash \text{Li}$	—

2 Exact enumeration and the complete spectrum

Definition 2.1. A matrix composition of mass n and height k is a $k \times C$ matrix of nonnegative integers with positive column sums and total entry sum n . Its columns and rows are ordered. Let $A(n, k)$ count these matrices, and let $T(n, k)$ count those with every row nonzero.

The word interpretation in A261781 is equivalent: a column of sum i specifies the multiplicities of the k letters in an alphabetically ordered word of length i . Every letter occurs somewhere precisely when every row is nonzero. For example, $T(2, 2) = 3$: the one-column matrix with entries 1, 1, and the two 2×2 permutation matrices.

2.1 Generating functions and a positive transform

Let $u > 0$ mark columns. Define

$$A(n, k; u) = \sum_B u^{C(B)}, \quad T(n, k; u) = \sum_{B: M(B)=0} u^{C(B)}.$$

At $u = 1$ these reduce to the unweighted counts. A nonzero column has generating function $(1 - z)^{-k} - 1$, since a column of sum i has $\binom{i+k-1}{i}$ possibilities. Taking an ordered sequence of

columns gives the classical identity

$$F_k(z; u) := \sum_{n \geq 0} A(n, k; u) z^n = \frac{1}{1 - u((1 - z)^{-k} - 1)} = \frac{(1 - z)^k}{(1 + u)(1 - z)^k - u}. \quad (2.1)$$

This also gives the positive recurrence

$$A(n, k; u) = u \sum_{i=1}^n \binom{i+k-1}{i} A(n-i, k; u) \quad (n \geq 1, k \geq 1). \quad (2.2)$$

Deleting any designated set of zero rows proves inclusion–exclusion:

$$T(n, k; u) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} A(n, j; u). \quad (2.3)$$

Equations (2.1)–(2.3) are established enumeration, not new formulas [4, 1, 2].

For later use let $F_q = \sum_{b=0}^q b! \left\{ \begin{smallmatrix} q \\ b \end{smallmatrix} \right\}$ be the ordered Bell or Fubini number, with $F_0 = 1$. The two Stirling triangles satisfy

$$x(x+1) \cdots (x+n-1) = \sum_q \begin{bmatrix} n \\ q \end{bmatrix} x^q, \quad (e^z - 1)^k = k! \sum_{q \geq k} \left\{ \begin{smallmatrix} q \\ k \end{smallmatrix} \right\} \frac{z^q}{q!}.$$

Expanding (2.1) at $u = 1$ in powers of $(1 - z)^{-k}$ gives

$$A(n, k) = \frac{1}{2} \sum_{b \geq 0} 2^{-b} \binom{n+kb-1}{n} = \frac{1}{n!} \sum_{q=0}^n \begin{bmatrix} n \\ q \end{bmatrix} F_q k^q.$$

For $n > 0$ the $b = 0$ term is zero. The identity $F_q = \frac{1}{2} \sum_{b \geq 0} b^q 2^{-b}$ follows by comparing the exponential generating functions, with the usual $0^0 = 1$ convention. Applying the k th binomial difference to j^q yields

$$T(n, k) = \frac{k!}{n!} \sum_{q=k}^n \begin{bmatrix} n \\ q \end{bmatrix} \left\{ \begin{smallmatrix} q \\ k \end{smallmatrix} \right\} F_q. \quad (2.4)$$

This is Proposition 29 of [4]. Its nonnegative summands will allow controlled asymptotic replacement without alternating-sum cancellation. In particular $T(n, n) = F_n$.

2.2 Every pole survives

For fixed $u > 0$ put

$$b = \frac{u}{1+u}, \quad L_u = \log(1 + 1/u), \quad a_{j,u} = b^{1/j}, \quad \omega_j = e^{2\pi i/j}.$$

The denominator in $F_j(z; u)$ has simple zeros $z_{j,\ell} = 1 - a_{j,u} \omega_j^\ell$, $0 \leq \ell < j$. Their residues are $-a_{j,u} \omega_j^\ell / ((1+u)j)$. Consequently, for $n \geq 1$,

$$A(n, j; u) = \frac{a_{j,u}}{(1+u)j} \sum_{\ell=0}^{j-1} \omega_j^\ell (1 - a_{j,u} \omega_j^\ell)^{-n-1}. \quad (2.5)$$

The polynomial part of F_j is the constant $1/(1+u)$, so (2.5) is deliberately stated only for $n \geq 1$. Its unweighted specialization is the published Binet formula [4, Proposition 12].

Theorem 2.2 (Minimal recurrence and spectrum). *Fix $k \geq 1$ and $u > 0$, and put $D = \binom{k+1}{2}$. The reduced denominator of $\sum_{n \geq 0} T(n, k; u) z^n$ is*

$$Q_{k,u}(z) = \prod_{j=1}^k ((1+u)(1-z)^j - u). \quad (2.6)$$

Thus the minimal eventual homogeneous constant-coefficient recurrence has order exactly D , over $\mathbb{C}$ as well as over $\mathbb{R}$. Its characteristic polynomial is

$$P_{k,u}(x) = \prod_{j=1}^k ((1+u)(x-1)^j - ux^j). \quad (2.7)$$

All D characteristic roots are simple and nonzero. At $u = 1$ the polynomial is monic with integer coefficients.

Proof. Roots belonging to different factors in (2.6) cannot coincide: their distances from 1 are $b^{1/j}$, strictly increasing with j . Within each factor the roots are simple. At a root of the j th factor, only the j th term in (2.3) has a pole, and its coefficient $(-1)^{k-j} \binom{k}{j}$ and residue are nonzero. Therefore no pole cancels. This proves the exact reduced denominator and its degree.

If an eventual recurrence of smaller order existed, multiplying the generating function by its recurrence polynomial would give a polynomial; that recurrence polynomial would have to vanish at every one of these D poles. Its degree would be at least D , a contradiction. Reversing the factors in (2.6) gives (2.7); its leading coefficient is $(1+u) - u = 1$ in each factor. No pole is zero, since $b < 1$, and every corresponding characteristic root is finite and nonzero. $\square$

Remark 2.3 (Both manuscripts; the published part). [Added 3 October 2026, batch 85B.] Manuscript 01 proves Theorem 2.2 at $u = 1$ by the same argument: the roots of $2(1-z)^j - 1$ satisfy $|1-z| = 2^{-1/j}$, which determines j , so the root sets are disjoint; at a root only the j th inclusion–exclusion summand is singular, with a nonzero coefficient; and an eventual recurrence of order D' would give a denominator of degree at most D' . It writes the recurrence as $T(n, k) = -\sum_{i=1}^D q_i T(n-i, k)$ for $n > D$, with $Q_{k,1}(z) = \sum_{i=0}^D q_i z^i$ normalized by $q_0 = 1$. The weighted statement (every $u > 0$) and Remark 2.4 are manuscript 06's alone.

What is new in either manuscript is the *minimality* (no cancellation, so no shorter recurrence even over $\mathbb{C}$) and the indexing of Remark 2.4. The existence of a recurrence of order $\binom{k+1}{2}$ is Proposition 25 of [4] (printed page 16, with m rows there): its proof displays the generating function of the no-zero-row column as $x^m F_m(x)$ over $(1-2x)(1-4x+2x^2) \cdots (2(1-x)^m - 1)$ with $\deg F_m \leq \binom{m}{2}$, which is $Q_{k,1}$ in our letters. Munarini, Poneti and Rinaldi do not claim minimality. Their Remark 26 gives the cases of two and three rows; for $k = 2$ it is the example that manuscript 01 prints,

$$\sum_{n \geq 0} T(n, 1) z^n = \frac{z}{1-2z}, \quad \sum_{n \geq 0} T(n, 2) z^n = \frac{z^2(3-2z)}{1-6z+10z^2-4z^3}, \quad (2.8)$$

whence $T(n, 2) = 6T(n-1, 2) - 10T(n-2, 2) + 4T(n-3, 2)$ for $n \geq 4$. The recurrence fails at $n = 3$ with the zero initial terms: there $T(3, 2) - 6T(2, 2) = 16 - 18 = -2 = (-1)^{k+D} 2^{k-1}$, the impulse of Remark 2.4 at $k = 2$, $D = 3$. Manuscript 01 presented its theorem as a proof of the OEIS order conjecture without citing Proposition 25; Section 11.2 records the correction.

Remark 2.4 (The initial impulse). The rational function for $T(n, k; u)$ tends to $(-1)^k u / (1+u)$ as $z \rightarrow \infty$: the $j = 0$ term of inclusion–exclusion is $(-1)^k$, and every other F_j tends to $1/(1+u)$. Hence its numerator has degree D , not less than D . The denominator recurrence begins at $n = D + 1$ for

the full column indexed from zero; it is an order- D recurrence on the tail $T(1, k; u), T(2, k; u), \dots$. At $u = 1$, writing $Q_{k,1}(z) = \sum_{i=0}^D q_i z^i$, one has

$$\sum_{i=0}^D q_i T(D-i, k) = (-1)^{k+D} 2^{k-1} \neq 0.$$

If a convention insists on an order- r recurrence starting already at index r on the unshifted column, its minimum is $D+1$, obtained by adjoining a zero characteristic root. This explains why the word “eventual” is part of the theorem.

2.3 The growth-constant conjecture

The smallest-modulus pole of $F_k(z; 1)$ is $r_k = 1 - 2^{-1/k}$, uniquely. It is also the smallest-modulus pole across all terms of (2.3), since $r_j > r_k$ for $j < k$. Thus, for each fixed k ,

$$T(n, k) \sim c_k d_k^n, \quad d_k = \frac{1}{1 - 2^{-1/k}}, \quad c_k = \frac{1}{2k(2^{1/k} - 1)}. \quad (2.9)$$

The underlying pole and unrestricted asymptotic are classical [4, Proposition 13]; inclusion–exclusion transfers them to T . These constants agree with the numerical columns in A261781.

Corollary 2.5. *The second conjecture recorded in A261781 holds:*

$$d_{k+1} - d_k \longrightarrow \frac{1}{L}.$$

More precisely,

$$d_k = \frac{k}{L} + \frac{1}{2} + \frac{L}{12k} - \frac{L^3}{720k^3} + O(k^{-5}), \quad (2.10)$$

$$d_{k+1} - d_k = \frac{1}{L} - \frac{L}{12k(k+1)} + O(k^{-4}), \quad (2.11)$$

$$c_k = \frac{1}{2L} - \frac{1}{4k} + \frac{L}{24k^2} - \frac{L^3}{1440k^4} + O(k^{-6}). \quad (2.12)$$

Proof. Use the Taylor expansion at zero of $1/(1 - e^{-x}) - 1/x$ with $x = L/k$. It is analytic in a neighborhood of zero; Taylor’s theorem supplies the stated remainders. Then subtract adjacent values and use $c_k = (d_k - 1)/(2k)$. $\square$

[Added 3 October 2026, batch 85B: manuscript 01 proves (2.9) and the expansions (2.10) and (2.12) in the same way, by expanding $x/(e^x - 1)$, and states, like manuscript 06, that the limit is immediate from the closed form of [4, Proposition 13]. The increment expansion (2.11) is manuscript 06’s alone. Manuscript 01 adds that $d_{k+1} - d_k$ *increases strictly* to $1/L$ and stays below it (Theorem 12.1).]

3 A factored Hankel determinant and its polynomial refinement

Let $T_u(n, k)$ be the polynomial that counts k -row matrix compositions of mass n , with no zero row or zero column, assigning weight u to each column. Thus $T_1(n, k) = T(n, k)$ is the triangle A261781. For $k \geq 1$ we set $T_u(0, k) = 0$. In the proofs we initially take $u > 0$; the resulting determinant identity is a polynomial identity and hence extends to every complex u .

Put

$$a = 1 + u, \quad b = u, \quad d = \binom{k+1}{2}, \quad c = \binom{k+1}{3}.$$

For $s \geq 1$ define the d -by- d Hankel determinant

$$H_{k,s}(u) = \det[T_u(s+p+q, k)]_{p,q=0}^{d-1}.$$

Theorem 3.1 (Maximal Hankel determinant). *For all integers $k, s \geq 1$, the following identity holds in $\mathbb{Z}[u]$:*

$$H_{k,s}(u) = u^{d+2c} (1+u)^{k(s-1)+2c} \prod_{j=1}^k \binom{k}{j}^j \prod_{1 \leq i < j \leq k} \left((1+u)^{(j-i)/g_{ij}} - u^{(j-i)/g_{ij}} \right)^{2g_{ij}}, \quad (3.1)$$

$$g_{ij} = \gcd(i, j).$$

Here $d + 2c = \sum_{j=1}^k j^2 = k(k+1)(2k+1)/6$. In particular, $H_{k,s}(u) > 0$ for every real $u > 0$, and

$$H_{k,s+1}(u) = (1+u)^k H_{k,s}(u).$$

The determinant is maximal for the positive-index tail: the infinite matrix $[T_u(1+p+q, k)]_{p,q \geq 0}$ has rank exactly d for $u > 0$. Every minor of order greater than d in that infinite matrix vanishes. The initial term at $n = 0$ requires separate treatment, given below.

3.1 Exact spectrum and residue weights

Use the column-weight notation $T_u(n, k) = T(n, k; u)$ from Section 2. The generating function and inclusion–exclusion have already been proved in (2.1)–(2.3).

For $j \geq 1$, set

$$P_j(t) = a(t-1)^j - bt^j.$$

Since $a - b = 1$, this is a monic polynomial of degree j . Its roots can be written

$$\lambda = \frac{1}{1-r}, \quad r^j = \frac{b}{a}.$$

For $u > 0$ all such roots are finite and nonzero. They are simple, and the root sets for distinct j are disjoint: a common root would give an r with two distinct prescribed moduli $(b/a)^{1/j}$. There are therefore exactly d roots in the union of the root sets.

Partial fractions yield, for $n \geq 1$,

$$T_u(n, k) = \sum_{j=1}^k \sum_{P_j(\lambda)=0} w_{j,\lambda} \lambda^n, \quad w_{j,\lambda} = (-1)^{k-j} \binom{k}{j} \frac{\lambda - 1}{aj}. \quad (3.2)$$

All the weights are nonzero. For completeness, at a pole $z_0 = 1/\lambda$, with $r = 1 - z_0$, the numerator of the rational generating function is b/a and the derivative of its denominator is $-ajr^{j-1}$. Its coefficient of $(1 - \lambda z)^{-1}$ is thus

$$\frac{-\lambda(b/a)}{-ajr^{j-1}} = \frac{\lambda r}{aj} = \frac{\lambda - 1}{aj}.$$

The polynomial part of the unrestricted generating function is the constant $1/a$, so it does not affect any $n \geq 1$ coefficient.

3.2 Root products, discriminants, and resultants

The following identities are enough to evaluate the determinant:

$$\prod_{P_j(\lambda)=0} \lambda = a, \quad (3.3)$$

$$\prod_{P_j(\lambda)=0} (\lambda - 1) = (-1)^{j+1} b, \quad (3.4)$$

$$\text{Disc}(P_j) = (-1)^{(j-1)(j-2)/2} j^j (ab)^{j-1}, \quad (3.5)$$

$$\text{Res}(P_i, P_j) = (-1)^i (ab)^i \left(a^{(j-i)/g} - b^{(j-i)/g} \right)^g, \quad 1 \leq i < j, \quad g = \gcd(i, j). \quad (3.6)$$

The first two identities follow from the constant coefficient of P_j and from $P_j(1) = -b$. At a root λ ,

$$P'_j(\lambda) = j \left(a(\lambda - 1)^{j-1} - b\lambda^{j-1} \right) = \frac{jb\lambda^{j-1}}{\lambda - 1}.$$

Taking products and using the root products gives

$$\prod_{P_j(\lambda)=0} P'_j(\lambda) = (-1)^{j+1} j^j (ab)^{j-1}.$$

For a monic degree- j polynomial, multiplication by $(-1)^{j(j-1)/2}$ converts this product into its discriminant. Since

$$\frac{j(j-1)}{2} + j + 1 \equiv \frac{(j-1)(j-2)}{2} \pmod{2},$$

this proves (3.5).

To prove the resultant formula, put $h = j - i$ and $g = \gcd(i, h)$. For a root of P_i write $r = 1 - 1/\lambda$, so that $r^i = b/a$. Then

$$P_j(\lambda) = \lambda^j (ar^j - b) = b\lambda^j (r^h - 1).$$

As r ranges through the i roots of $r^i = b/a$, its h th powers are the i/g roots of

$$v^{i/g} = (b/a)^{h/g},$$

each occurring with multiplicity g . Consequently

$$\begin{aligned} \prod_{r^i=b/a} (r^h - 1) &= (-1)^i \left(1 - (b/a)^{h/g} \right)^g \\ &= (-1)^i a^{-h} \left(a^{h/g} - b^{h/g} \right)^g. \end{aligned}$$

Because P_i is monic, its resultant with P_j is the product of $P_j(\lambda)$ over its roots. Using $\prod \lambda = a$ gives

$$\begin{aligned} \text{Res}(P_i, P_j) &= b^i a^j (-1)^i a^{-h} \left(a^{h/g} - b^{h/g} \right)^g \\ &= (-1)^i (ab)^i \left(a^{h/g} - b^{h/g} \right)^g, \end{aligned}$$

as asserted.

3.3 Vandermonde evaluation

Enumerate all d roots in (3.2) as $\lambda_1, \dots, \lambda_d$, keeping the j -blocks together, and write w_ν for the corresponding weights. The Hankel matrix factors as

$$[T_u(s + p + q, k)]_{p,q=0}^{d-1} = V \operatorname{diag}(w_\nu \lambda_\nu^s)_{\nu=1}^d V^\top, \quad V_{p,\nu} = \lambda_\nu^p.$$

Hence

$$H_{k,s}(u) = \left(\prod_{\nu=1}^d w_\nu \lambda_\nu^s \right) \prod_{\mu < \nu} (\lambda_\nu - \lambda_\mu)^2. \quad (3.7)$$

Splitting the squared Vandermonde product into within-block and between-block factors gives

$$\prod_{\mu < \nu} (\lambda_\nu - \lambda_\mu)^2 = \prod_{j=1}^k \operatorname{Disc}(P_j) \prod_{1 \leq i < j \leq k} \operatorname{Res}(P_i, P_j)^2.$$

For one block, the weights, the root power, and the discriminant combine to

$$\begin{aligned} & \left(\prod_{P_j(\lambda)=0} w_{j,\lambda} \lambda^s \right) \operatorname{Disc}(P_j) \\ &= (-1)^{j(k-j)+\binom{j}{2}} \binom{k}{j}^j b^j a^{s-1}. \end{aligned}$$

There is no residual sign after multiplying over j , because

$$\sum_{j=1}^k \left(j(k-j) + \binom{j}{2} \right) = 2 \binom{k+1}{3} = 2c.$$

The squared resultants supply

$$(ab)^{2 \sum_{1 \leq i < j \leq k} i} \prod_{i < j} \left(a^{(j-i)/g_{ij}} - b^{(j-i)/g_{ij}} \right)^{2g_{ij}} = (ab)^{2c} \prod_{i < j} \left(a^{(j-i)/g_{ij}} - b^{(j-i)/g_{ij}} \right)^{2g_{ij}}.$$

Substitution in (3.7) proves (3.1) for $u > 0$. Both sides are polynomials in u , so agreement on this interval proves the identity in $\mathbb{Z}[u]$.

3.4 The unweighted specialization

At $u = 1$ the formula becomes

$$\det[T(s + p + q, k)]_{p,q=0}^{d-1} = 2^{k(s-1)+2\binom{k+1}{3}} \prod_{j=1}^k \binom{k}{j}^j \prod_{1 \leq i < j \leq k} \left(2^{(j-i)/\gcd(i,j)} - 1 \right)^{2\gcd(i,j)}. \quad (3.8)$$

The first few values of $H_{k,1}(1)$ are

$$H_{1,1}(1) = 1, \quad H_{2,1}(1) = 8, \quad H_{3,1}(1) = 62208.$$

For example, the complete $k = 3$ polynomial is

$$H_{3,s}(u) = 27u^{14}(1+u)^{3s+5}(1+2u)^2.$$

This extends the Cassini-type phenomenon already observed for unrestricted matrix compositions. In [4, Proposition 21], the unrestricted height- k sequence has a k -by- k determinant $(-1)^{\lfloor k/2 \rfloor} 2^{s-1}$. Enforcing all rows to be used mixes the root sets for heights $1, \dots, k$. The determinant above has order $d = k(k+1)/2$ and includes the explicit pairwise resultant factors caused by that mixture. The primary paper does not state this order- d determinant; the proof here is independent of any priority claim.

3.5 Recurrence minimality and the initial-term correction

The nonzero determinant is an independent proof of the exact tail rank d and the minimality in Theorem 2.2. It also makes the initial impulse of Remark 2.4 quantitative.

At $n = 0$ the spectral sum has the value

$$\sum_{\nu=1}^d w_{\nu} = (-1)^{k+1} \frac{u}{1+u},$$

whereas $T_u(0, k) = 0$. Thus the all-index identity is

$$T_u(n, k) = \sum_{\nu=1}^d w_{\nu} \lambda_{\nu}^n + (-1)^k \frac{u}{1+u} \mathbf{1}_{\{n=0\}}. \quad (3.9)$$

In particular, the minimal forward-shift annihilator valid from $n = 0$ is $tP_{k,u}(t)$; the extra root is zero. In the conventional eventual-recurrence language the order is still d . This distinction also accounts for the rank of the unshifted infinite Hankel matrix when $u > 0$:

$$\text{rank}[T_u(p+q, k)]_{p,q \geq 0} = d+1.$$

Indeed, adjoining the zero root to the Vandermonde factorization gives the exact exceptional determinant

$$\det[T_u(p+q, k)]_{p,q=0}^d = (-1)^k \frac{u}{1+u} H_{k,2}(u). \quad (3.10)$$

The extra squared Vandermonde factors are $\prod_{\nu} \lambda_{\nu}^2$, which change $H_{k,0}$ for the pure spectral extension into $H_{k,2}$. At $u = 1$,

$$\det[T(p+q, k)]_{p,q=0}^d = (-1)^k 2^{k-1} H_{k,1}(1).$$

3.6 Coefficient positivity and the location of the zeros

Every nonconstant difference factor on the right of (3.1) is

$$(1+u)^m - u^m = \sum_{r=0}^{m-1} \binom{m}{r} u^r,$$

which has positive coefficients. Therefore $H_{k,s}(u) \in \mathbb{Z}_{\geq 0}[u]$. This statement concerns these maximal determinants. It does not assert total positivity of the Hankel matrix: for $k = 2$, for example,

$$\det \begin{pmatrix} T_u(1, 2) & T_u(2, 2) \\ T_u(2, 2) & T_u(3, 2) \end{pmatrix} = -u^2(1+2u)^2 < 0 \quad (u > 0).$$

The factorization also determines the geometry of all complex zeros. Apart from $u = 0$ and $u = -1$, every zero lies on the vertical line $\text{Re } u = -1/2$. Indeed, a zero of $(1+u)^m - u^m$ has the form

$$u = \frac{1}{\zeta - 1}, \quad \zeta^m = 1, \quad \zeta \neq 1,$$

and

$$\frac{1}{e^{i\theta} - 1} = -\frac{1}{2} - \frac{i}{2} \cot \frac{\theta}{2}.$$

The multiplicity at $u = 0$ is exactly $\sum_{j=1}^k j^2$. When its displayed exponent is positive, the multiplicity at $u = -1$ is exactly $k(s-1) + 2\binom{k+1}{3}$. All other multiplicities follow by collecting the cyclotomic difference factors in the product.

4 An exponential error bound uniform in the row count

The essential analytic input for the growing-row results is stronger than a fixed-parameter pole asymptotic. The bound below does not deteriorate when arbitrarily many poles are present.

Theorem 4.1 (Uniform dominant-pole estimate). *For every $n, k \geq 1$,*

$$A(n, k) = c_k d_k^n (1 + \varepsilon_{n,k}), \quad |\varepsilon_{n,k}| \leq B \rho^{n-1}, \quad (4.1)$$

where

$$B = \frac{\pi^2 L^2}{24}, \quad \rho = \left(1 + \frac{8}{L^2}\right)^{-1/2} < 0.239. \quad (4.2)$$

More generally, with $b = u/(1+u)$, $L_u = \log(1+1/u)$, $a_{k,u} = e^{-L_u/k}$, $r_{k,u} = 1 - a_{k,u}$, and

$$d_k(u) = r_{k,u}^{-1}, \quad \mu_k(u) = \frac{a_{k,u}}{(1+u)kr_{k,u}},$$

one has

$$A(n, k; u) = \mu_k(u) d_k(u)^n (1 + \varepsilon_{n,k}(u)), \quad |\varepsilon_{n,k}(u)| \leq B(u) \rho(u)^{n-1}, \quad (4.3)$$

where

$$B(u) = \frac{\pi^2 L_u^2}{48b}, \quad \rho(u) = \left(1 + \frac{16b}{L_u^2}\right)^{-1/2}.$$

For u in a fixed compact subset of $(0, \infty)$ the constants can be replaced by common $B_I < \infty$ and $\rho_I < 1$.

Proof. In (2.5), divide by the positive $\ell = 0$ term. Its coefficient is exactly $\mu_k(u) d_k(u)^n$. If $j = \min(\ell, k - \ell)$, then $\sin(\pi\ell/k) \geq 2j/k$, $a_{k,u} \geq b$, and $r_{k,u} \leq L_u/k$. Therefore

$$\begin{aligned} |1 - a_{k,u} \omega_k^\ell|^2 &= r_{k,u}^2 + 4a_{k,u} \sin^2(\pi\ell/k) \\ &\geq r_{k,u}^2 \left(1 + \frac{16b}{L_u^2} j^2\right). \end{aligned}$$

Put $D_u = 16b/L_u^2$. There are at most two roots for each j , so

$$\begin{aligned} |\varepsilon_{n,k}(u)| &\leq 2 \sum_{j \geq 1} (1 + D_u j^2)^{-(n+1)/2} \\ &\leq 2(1 + D_u)^{-(n-1)/2} \sum_{j \geq 1} \frac{1}{D_u j^2} = \frac{\pi^2}{3D_u} \rho(u)^{n-1}. \end{aligned}$$

For $k = 1$ the omitted sum is empty and the error is exactly zero. On a compact interval use its minimum b and maximum L_u in the same inequalities. At $u = 1$, $b = 1/2$, giving (4.2). $\square$

Remark 4.2 (A second route with a k -dependent constant). [Added 3 October 2026, batch 85B.] At $u = 1$ manuscript 01 proves the same one-pole dominance with a different constant:

$$|\varepsilon_{n,k}| \leq (k-1) \rho^{n+1} \quad (n, k \geq 1), \quad (4.4)$$

with the same $\rho = L/\sqrt{L^2 + 8}$ (manuscript 01 calls it η ; see Table 2). Its proof bounds each of the $k-1$ nonprincipal terms of (2.5) separately: with $a_k \geq \frac{1}{2}$, $r_k \leq L/k$ and $\sin(\pi/k) \geq 2/k$ one gets $r_k/|1 - a_k \omega_k^\ell| \leq \rho$ for every $\ell \neq 0$, and each ratio is at most ρ^{n+1} . Since $\rho^2 = L^2/(L^2 + 8) < 0.057$ and $B = \pi^2 L^2/24 > 0.197$, the bound (4.4) is the sharper one for $k \leq 4$ and Theorem 4.1 for $k \geq 5$; only Theorem 4.1 is uniform in k , which the growing-row results need. Manuscript 01 says so itself: its relative bound is useful only when $k\rho^n$ is small, and is not claimed to be small for large k and fixed n ; both of its growing regimes satisfy the condition by a wide margin.]

Remark 4.3. The theorem controls the unrestricted count $A(n, k)$ uniformly. It does not say that $T(n, k) \sim c_k d_k^n$ uniformly: the subtraction enforcing all rows nonzero becomes significant when $n \approx k \log k$ and exponentially significant when $n \asymp k$. The next sections quantify these effects.

5 The empty-row transition and its first correction

Put $L = \log 2$ and suppose that $A(n, k)$ counts unrestricted objects on k labelled rows, with

$$\sum_{n \geq 0} A(n, k) z^n = \frac{(1 - z)^k}{2(1 - z)^k - 1}.$$

For $m \geq 1$ write

$$d_m = (1 - e^{-L/m})^{-1}, \quad c_m = \frac{d_m - 1}{2m}.$$

Theorem 4.1 supplies the estimate

$$A(n, m) = c_m d_m^n (1 + \varepsilon_{n,m}), \quad |\varepsilon_{n,m}| \leq C_0 \rho^{n-1}, \quad 0 < \rho < 1, \quad (5.1)$$

uniformly for $m \geq 1$ and $n \geq 2$.

Choose an unrestricted matrix of mass n and height k uniformly and let M be its number of unused rows. Its factorial moments are exactly

$$b_j := \mathbb{E}[(M)_j] = (k)_j \frac{A(n, k-j)}{A(n, k)} \quad (0 \leq j \leq k). \quad (5.2)$$

Here $(x)_j = x(x-1) \cdots (x-j+1)$, $b_0 = 1$, and $b_k = 0$ when $n > 0$.

Theorem 5.1 (Coverage-window expansion). *Fix $S, R < \infty$. Set*

$$h = \frac{n}{k}, \quad s = h - \log k, \quad \eta = k e^{-h} = e^{-s}.$$

As $k \rightarrow \infty$, uniformly for integers n satisfying $|s| \leq S$ and complex w satisfying $|w| \leq R$,

$$\mathbb{E}(1+w)^M = e^{\eta w} \left\{ 1 - \frac{(1+h)\eta^2 w^2 + (1-L)h\eta w}{2k} + O_{S,R} \left(\frac{h^2}{k^2} \right) \right\}. \quad (5.3)$$

In particular,

$$\mathbb{P}(M=0) = e^{-\eta} \left\{ 1 + \frac{(1-L)h\eta - (1+h)\eta^2}{2k} + O_S \left(\frac{(\log k)^2}{k^2} \right) \right\}. \quad (5.4)$$

The definition $s = n/k - \log k$ includes the integer rounding of n ; there is no unstated choice of rounding convention.

Proof. We first obtain a bound valid for all inclusion–exclusion indices, not only fixed j . The elementary inequalities

$$\frac{m}{L} \leq d_m \leq \frac{m}{L} + 1$$

follow respectively from $1 - e^{-x} \leq x$ and $e^x \geq 1 + x$. Also, c_m is bounded above and bounded away from zero uniformly for $m \geq 1$. Equation (5.1) therefore gives, for all sufficiently large k in the indicated window and $0 \leq j \leq k-1$,

$$b_j \leq C(k)_j \left(\frac{d_{k-j}}{d_k} \right)^n$$

$$\begin{aligned}
&\leq Ck^j \left(1 - \frac{j-L}{k}\right)^n \\
&\leq Ce^{Lh}(ke^{-h})^j = Ce^{Lh}\eta^j.
\end{aligned} \tag{5.5}$$

The same bound holds for $j = k$ since $b_k = 0$.

Let $J = \lceil 4 \log k \rceil$. Since $\eta \leq e^S$ and $e^{Lh} \leq e^{LS}k^L$, the factorial tail bound gives

$$\sum_{j>J} \frac{|w|^j b_j}{j!} \leq Ck^L \sum_{j>J} \frac{(Re^S)^j}{j!} = O_{S,R,B}(k^{-B}) \tag{5.6}$$

for every fixed $B > 0$. Indeed the logarithm of the last factorial tail is at most $-J \log J + O_{S,R}(J)$. This bound directly justifies absolute truncation of the exact factorial-moment polynomial.

For the remaining indices $1 \leq j \leq J$, the needed expansion is uniform. Define

$$g_0(x) = \log \frac{Lx/2}{\sinh(Lx/2)}, \quad g_0(0) = 0.$$

Near zero, g_0 is analytic, $g_0(x) = O(x^2)$ and $g'_0(x) = O(x)$. The exact identities

$$\log d_m = \log \frac{m}{L} + \frac{L}{2m} + g_0(1/m), \quad \log c_m = -\log(2L) - \frac{L}{2m} + g_0(1/m)$$

yield

$$\begin{aligned}
\log \frac{d_{k-j}}{d_k} &= -\frac{j}{k} - \frac{j^2}{2k^2} + \frac{Lj}{2k^2} + O\left(\frac{j^3}{k^3}\right), \\
\log \frac{c_{k-j}}{c_k} &= O\left(\frac{j}{k^2}\right), \\
\log(k)_j &= j \log k - \frac{j(j-1)}{2k} + O\left(\frac{j^3}{k^2}\right).
\end{aligned}$$

All constants here are absolute, once $J \leq k/2$. Consequently

$$\log b_j = j \log \eta + \frac{-j(j-1) + h(Lj - j^2)}{2k} + O\left(\frac{(1+h)j^3}{k^2} + \rho^{n-1}\right). \tag{5.7}$$

Since $hJ^2/k \rightarrow 0$, exponentiation is uniform and gives

$$b_j = \eta^j \left\{ 1 + \frac{-j(j-1) + h(Lj - j^2)}{2k} + O\left(\frac{h^2 j^4}{k^2} + \rho^{n-1}\right) \right\}. \tag{5.8}$$

The index $j = 0$ is exact and is treated separately.

Now use the exact identity

$$\mathbb{E}(1+w)^M = \sum_{j=0}^k \frac{b_j w^j}{j!}.$$

The error in (5.8) sums to $O_{S,R}(h^2/k^2)$, because $\sum_{j \geq 0} j^4 (Re^S)^j / j! < \infty$ and ρ^{n-1} is smaller than every fixed power of k^{-1} in this window. Equation (5.6) permits extension of all the main sums to infinity. With $x = \eta w$,

$$\sum_{j \geq 0} \frac{[-j(j-1) + h(Lj - j^2)]x^j}{j!} = -[(1+h)x^2 + (1-L)hx]e^x.$$

This proves (5.3); the additive and relative forms of its error are equivalent on each fixed disk because $|e^{\eta w}|$ is bounded away from zero there. Set $w = -1$, with the polynomial interpretation $0^0 = 1$, to obtain (5.4). $\square$

[Added 3 October 2026, batch 85B: manuscript 01 proves the case $w = -1$, that is (5.4) with error $O((1+h)^2/k^2)$, which is the same order, and the Poisson limit of M , by a second route. It expands the factorial moments exactly as in (5.8) (its form is $\log(\mathbb{E}(M)_j/\eta^j) = -(h+1)j^2 + (Lh+1)j/(2k) + O((1+h)(1+j)^4/k^2)$ for $j \leq C \log k$, the same polynomial), then truncates the inclusion–exclusion series for $\mathbb{P}(M=0)$ by the Bonferroni inequalities after $R \asymp \log k$ terms, the next term being at most $C^{R+1}/(R+1)!$, and sums with $\sum_j (-\eta)^j j/j! = -\eta e^{-\eta}$ and $\sum_j (-\eta)^j j^2/j! = (\eta^2 - \eta)e^{-\eta}$. The same device gives every fixed order (Theorem 13.1). Manuscript 06 proves the whole generating function in complex w , and from it the probability-mass and total-variation statements below, which manuscript 01 does not have.]

5.1 A summable expansion of the entire missing-row law

For $\eta > 0$ put $\pi_\eta(r) = e^{-\eta}\eta^r/r!$ for $r \geq 0$, and set $\pi_\eta(r) = 0$ for negative integers r . Define

$$C_2(r; \eta) = r(r-1) - 2\eta r + \eta^2 = (r-\eta)^2 - r.$$

Corollary 5.2 (Probability-mass and total-variation expansion). *Under the assumptions of Theorem 5.1, uniformly for $r \geq 0$,*

$$\mathbb{P}(M=r) = \pi_\eta(r) \left\{ 1 + \frac{(1-L)h(\eta-r) - (1+h)C_2(r; \eta)}{2k} \right\} + \mathcal{R}_{k,r}, \quad (5.9)$$

where

$$|\mathcal{R}_{k,r}| \leq C_S 2^{-r} \frac{h^2}{k^2}, \quad \sum_{r \geq 0} |\mathcal{R}_{k,r}| = O_S \left(\frac{h^2}{k^2} \right).$$

In particular, if $X \sim \text{Poisson}(\eta)$, then

$$d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(\eta)) = \frac{1}{4k} \mathbb{E} |(1-L)h(\eta-X) - (1+h)C_2(X; \eta)| + O_S \left(\frac{h^2}{k^2} \right). \quad (5.10)$$

The total-variation distance is $\Theta_S((\log k)/k)$, uniformly for $|s| \leq S$. If $s \rightarrow c$ and $\eta_0 = e^{-c}$, its leading constant is given by

$$\frac{k}{\log k} d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(ke^{-n/k})) \longrightarrow \frac{1}{4} \mathbb{E} |(1-L)(\eta_0 - X_0) - C_2(X_0; \eta_0)| > 0, \quad (5.11)$$

where $X_0 \sim \text{Poisson}(\eta_0)$.

Proof. Substitute $w = z - 1$ in (5.3). Its error is uniformly $O_S(h^2/k^2)$ on $|z| \leq 2$, using $R = 3$. Cauchy's coefficient bound gives $|\mathcal{R}_{k,r}| \leq C_S 2^{-r} h^2/k^2$. The coefficients of $(z-1)e^{\eta(z-1)}$ and $(z-1)^2 e^{\eta(z-1)}$ are respectively

$$\pi_\eta(r-1) - \pi_\eta(r), \quad \pi_\eta(r-2) - 2\pi_\eta(r-1) + \pi_\eta(r).$$

Writing these as multiples of $\pi_\eta(r)$ proves (5.9). The reverse triangle inequality and summability of $\mathcal{R}_{k,r}$ prove (5.10).

The function

$$D(\eta) = \mathbb{E} |(1 - L)(\eta - X) - C_2(X; \eta)|, \quad X \sim \text{Poisson}(\eta),$$

is continuous for $\eta > 0$ and strictly positive: its integrand is the absolute value of a nonzero quadratic polynomial and a Poisson law with positive mean assigns positive mass to every nonnegative integer. Therefore D has a positive minimum and a finite maximum on $[e^{-S}, e^S]$. The difference between the expectation in (5.10) and $hD(\eta)$ is bounded by $\mathbb{E}|C_2(X; \eta)| = O_S(1)$. This proves the uniform two-sided rate and the claimed limit. $\square$

5.2 Mean and variance interpretation

The same expansion gives

$$\nu_k := \mathbb{E}M = \eta - \frac{(1 - L)h\eta}{2k} + O_S\left(\frac{h^2}{k^2}\right),$$

and

$$\text{Var}(M) - \mathbb{E}M = -\frac{(1 + h)\eta^2}{k} + O_S\left(\frac{h^2}{k^2}\right).$$

Thus the correction includes both a shift of the Poisson mean and a negative second factorial cumulant. Adjusting only the mean does not remove the leading dependence correction. More precisely,

$$d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(\nu_k)) = \frac{1 + h}{4k} \mathbb{E}|C_2(X; \eta)| + O_S\left(\frac{h^2}{k^2}\right), \quad X \sim \text{Poisson}(\eta).$$

To see this, replace the linear term in (5.3) by $e^{(\nu_k - \eta)w}$, with uniform error $O_S(h^2/k^2)$, and apply the same coefficient argument. Replacing η by ν_k in the expectation changes the displayed answer by only $O_S(h^2/k^2)$: the expectation is locally Lipschitz as a function of its positive Poisson parameter, by dominated summation of the polynomially weighted Poisson series on a compact parameter interval.

Corollary 5.3 (The coverage threshold). *If $n/k - \log k \rightarrow s \in \mathbb{R}$, then*

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \frac{T(n, k)}{A(n, k)} \rightarrow \exp(-e^{-s}).$$

For bounded $s_k = n/k - \log k$, with $\eta = ke^{-n/k}$ and $h = n/k$,

$$T(n, k) = c_k d_k^n e^{-\eta} \left[1 + \frac{(1 - L)h\eta - (1 + h)\eta^2}{2k} + O\left(\frac{(\log k)^2}{k^2}\right) \right]. \quad (5.12)$$

Proof. The generating-function expansion implies the Poisson distributional limit; alternatively use the summable probability-mass expansion. The event $M = 0$ is precisely membership in the class counted by T . Multiply its probability by Theorem 4.1; the exponential error is negligible at the displayed scale. $\square$

[Added 3 October 2026, batch 85B: the limit statements are also manuscript 01's (its Theorem on the Poisson transition), proved there by marked inclusion–exclusion with $\mathbb{E}(M)_{j+m}/(m!j!)$ for each fixed point mass m , or equivalently by the factorial-moment method. Table 8 in Part II gives manuscript 01's coverage probabilities as ratios of exact integers for $k = 10, \dots, 120$; at $k = 20$, $n = 60$ they agree with Table 3 in every printed digit.]

Remark 5.4 (Moving versus fixed Poisson mean). The sharp rate in (5.11) compares the actual law with $\text{Poisson}(ke^{-n/k})$. If only $n/k - \log k \rightarrow s$ is assumed, replacing this moving mean by e^{-s} incurs an additional error governed by the rate of that convergence. For $n = \text{round}(k(\log k + s))$, the additional error is $O(1/k)$ and does not change the leading $(\log k)/k$ constant. No such rate is asserted for an arbitrary unspecified $o(1)$ perturbation.

6 Column counts and joint Gaussian–Poisson independence

The row-coverage law survives a fixed positive weight on the number of columns. This permits a useful independence theorem by exponential tilting. Under the probability measure proportional to $u^{C(B)}$, denote expectation and probability by $\mathbb{E}_u$ and $\mathbb{P}_u$.

Lemma 6.1 (Uniformity under column weights). *If $n/k - \log k \rightarrow s \in \mathbb{R}$, then, uniformly for u in each fixed compact interval $I \subset (0, \infty)$,*

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \mathbb{P}_u(M = m) \rightarrow e^{-e^{-s}} \frac{e^{-ms}}{m!} \quad (m \geq 0 \text{ fixed}).$$

The same conclusion holds along any sequence of weights in I .

Proof. Deleting designated zero rows gives the exact identity

$$\mathbb{E}_u[(M)_j] = (k)_j \frac{A(n, k-j; u)}{A(n, k; u)}.$$

Use Theorem 4.1. Uniformly for $u \in I$, with $L_u = \log(1 + 1/u)$ and $q = 1/(1 + u)$,

$$\log d_k(u) = \log k - \log L_u + \frac{L_u}{2k} + O_I(k^{-2}), \quad \log \mu_k(u) = \log \frac{q}{L_u} - \frac{L_u}{2k} + O_I(k^{-2}).$$

For each fixed j this gives

$$\log \mathbb{E}_u[(M)_j] = -j(n/k - \log k) + O_{I,j}((\log k)/k).$$

Thus all fixed factorial moments tend uniformly to e^{-js} . To make the distributional step explicit, expand $\mathbb{E}_u z^M$ at $z = 1$ for $0 \leq z \leq 1$. Taylor's remainder after $J - 1$ terms is bounded by $\mathbb{E}_u[(M)_J]/J!$. First pass to the limit with J fixed, then let $J \rightarrow \infty$; the limiting error $e^{-Js}/J!$ tends to zero. This identifies the Poisson probability generating function. The first moment bounds give tightness. Every subsequential limit is therefore Poisson, uniformly in the allowed weights. On the nonnegative integers, weak convergence gives convergence of each point probability by taking an interval with half-integer endpoints. $\square$

Fix a baseline $u_0 > 0$. In the formulas below evaluate

$$b = \frac{u_0}{1 + u_0}, \quad q = \frac{1}{1 + u_0}, \quad L_{u_0} = \log(1 + 1/u_0), \quad a = e^{-L_{u_0}/k}, \quad r = 1 - a$$

at this baseline. Define

$$\mu_k(u_0) = \frac{qa}{kr}, \quad \sigma_k^2(u_0) = \frac{\mu_k(u_0)^2}{a} - b\mu_k(u_0) = \frac{qa}{k^2 r^2} (q - bkr). \quad (6.1)$$

These are the first two derivatives of $\ell_k(v) = \log d_k(u_0 e^v)$ at $v = 0$. Indeed logarithmic differentiation gives $q' = -bq$, $L'_u = -q$, $a' = aq/k$, and $r' = -aq/k$. The variance constant is strictly positive, since $q/b = e^{L_{u_0}} - 1 > L_{u_0} \geq kr$.

Theorem 6.2 (Joint limit and conditional Gaussian laws). *Suppose $n/k - \log k \rightarrow s \in \mathbb{R}$, and choose matrix compositions of mass n and height k with probabilities proportional to u_0^C , for fixed $u_0 > 0$. Then*

$$\left(\frac{C - n\mu_k(u_0)}{\sqrt{n\sigma_k^2(u_0)}}, M \right) \xrightarrow{d} (Z, P), \quad (6.2)$$

where Z is standard normal, P has law $\text{Poisson}(e^{-s})$, and Z, P are independent. For each fixed $m \geq 0$, the same Gaussian limit holds conditional on $M = m$. In particular it holds for the A261781 class, which requires $M = 0$.

Proof. Put $s_k = \sqrt{n\sigma_k^2(u_0)}$, $X = (C - n\mu_k(u_0))/s_k$, and $u_v = u_0 e^{v/s_k}$ for a fixed real v . The exact tilting identity is

$$\mathbb{E}_{u_0} e^{vX} = e^{-vn\mu_k(u_0)/s_k} \frac{A(n, k; u_v)}{A(n, k; u_0)}.$$

The two weights lie in a common compact interval. Applying (4.3) and Taylor's theorem gives

$$\begin{aligned} \log \mathbb{E}_{u_0} e^{vX} &= n [\ell_k(v/s_k) - \ell_k(0) - \mu_k(u_0)v/s_k] + \log \frac{\mu_k(u_v)}{\mu_k(u_0)} + O(\rho_I^{n-1}) \\ &= \frac{v^2}{2} + O_v(n^{-1/2}) + O(\rho_I^{n-1}). \end{aligned}$$

The derivatives used here are bounded uniformly in k : write

$$\log d_k(u) = \log k - \log L_u + g(L_u/k), \quad g(x) = \log \frac{x}{1 - e^{-x}},$$

where g is smooth at zero. The variance constants have a positive limit and remain bounded away from zero and infinity. The logarithmic derivatives of μ_k are likewise uniformly bounded. Convergence of moment generating functions on a neighborhood of zero proves the marginal normal limit.

For every fixed m , the conditional tilting identity is also exact:

$$\mathbb{E}_{u_0} [e^{vX} \mid M = m] = \mathbb{E}_{u_0} e^{vX} \frac{\mathbb{P}_{u_v}(M = m)}{\mathbb{P}_{u_0}(M = m)}.$$

Both probabilities tend to the same strictly positive Poisson probability by Lemma 6.1. Their ratio tends to one, proving the conditional Gaussian limit. Summing the conditional limits over finitely many m and controlling the remaining mass by tightness of M proves the joint limit and independence. $\square$

For the unweighted model these constants simplify to

$$\mu_k = c_k, \quad \sigma_k^2 = \frac{a_k(1 - kr_k)}{4k^2r_k^2}, \quad \mu_\infty = \frac{1}{2L}, \quad \sigma_\infty^2 = \frac{1 - L}{4L^2}. \quad (6.3)$$

In the coverage window, using $n\mu_\infty$ instead of $n\mu_k$ also gives the same limit: their difference is $O(n/k) = O(\log k)$, which is $o(\sqrt{n})$. The finite- k centering keeps this correction visible. Louchard's fixed-row Gaussian theorem is an antecedent [6]; the new conclusion here is uniformity with growing rows and its joint and conditional form.

7 Uniform asymptotics on dense rays

We now require every row to be nonzero and let n/k stay bounded above and away from 1. The Poisson-window expansion does not apply here. The positive transform (2.4) gives a different route.

7.1 The saddle and the first correction

Put $f(z) = e^z - 1$. For $\lambda > 1$, let $r = r(\lambda) > 0$ solve

$$\frac{r}{1 - e^{-r}} = \lambda. \quad (7.1)$$

With $\mathcal{D} = r \, d/dr$, define

$$\kappa_j = \mathcal{D}^j \log f(r), \quad b = \kappa_2 = \lambda(1 + r - \lambda), \quad v = \frac{Lr}{2}. \quad (7.2)$$

In particular $\kappa_1 = \lambda$ and $b > 0$.

Theorem 7.1 (Uniform ray expansion). *Let K be any fixed compact subset of $(1, \infty)$. Uniformly over integers $n, k \rightarrow \infty$ with $\lambda = n/k \in K$,*

$$T(n, k) = \frac{n! f(r)^k e^v}{2^{L^{n+1}} r^n \sqrt{2\pi k b}} \left[1 + \frac{E_1(\lambda)}{k} + O_K(k^{-2}) \right], \quad (7.3)$$

where

$$E_1(\lambda) = \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v + v^2}{2b} + \frac{v^2/3 - v}{\lambda}. \quad (7.4)$$

The expansion continues to every fixed order in integral powers of $1/k$, uniformly on K . Section 7.4 gives an explicit finite coefficient-generation rule at each order.

The derivative definitions in (7.2) are exact finite formulas, not additional implicitly determined constants. If expanded,

$$\begin{aligned} \kappa_3 &= \lambda(2\lambda^2 - 3\lambda r - 3\lambda + r^2 + 3r + 1), \\ \kappa_4 &= \lambda(-6\lambda^3 + 12\lambda^2 r + 12\lambda^2 - 7\lambda r^2 - 18\lambda r - 7\lambda \\ &\quad + r^3 + 6r^2 + 7r + 1). \end{aligned}$$

7.2 Reducing the positive sum to an analytic amplitude

The Fubini generating function is

$$\sum_{q \geq 0} F_q \frac{z^q}{q!} = \frac{1}{2 - e^z}.$$

Its poles are $L + 2\pi i j$, $j \in \mathbb{Z}$, with residue $-1/2$. Subtracting the pole at L and applying Cauchy's coefficient estimate on a circle between it and the next poles proves

$$F_q = \frac{q!}{2^{L^{q+1}}} (1 + O(\vartheta^q)) \quad (7.5)$$

for any fixed ϑ strictly between $L/\sqrt{L^2 + 4\pi^2}$ and 1. Every term in (2.4) is nonnegative and $q \geq k$, so this replacement incurs a relative error $O(\vartheta^k)$ in the entire sum.

[Added 3 October 2026, batch 85B: manuscript 01 proves the same estimate (its Lemma on the Fubini approximation) by subtracting $1/(2(L-z))$ from $1/(2-e^z)$ and applying Cauchy's bound on a circle of radius between L/ϑ and $\sqrt{L^2 + 4\pi^2}$. Both are classical. The repository's transseries volume has the exact convergent pole-lattice formula $F_q = \frac{q!}{2} \sum_{j \in \mathbb{Z}} (L + 2\pi i j)^{-q-1}$, $q \geq 1$ (Theorem `q2:thm:fubini` of [11]), of which (7.5) keeps the term $j = 0$; and the generating function $1/(2-e^z)$ is formalized in Lean as `Fabius.two_sub_exp_mul_egfA_fubini` [12].]

Define the polynomial

$$H_{n,k}(z) = \sum_{d=0}^{n-k} L^d \begin{bmatrix} n \\ n-d \end{bmatrix} \left(\frac{(n-d)!}{n!} \right)^2 z^d. \quad (7.6)$$

Using $[z^q]f(z)^k = k! \binom{q}{k} / q!$, including both factorials from (2.4) and (7.5), gives

$$T(n, k) = \frac{n!}{2L^{n+1}} [z^n] f(z)^k H_{n,k}(z) (1 + O(\vartheta^k)). \quad (7.7)$$

Terms with $d > n - k$ would have zero coefficient contribution since $f(z)^k$ begins at degree k . Notice that the numerator in this formula is $n!$, not $1/n!$; the square of the factorial ratio in (7.6) is essential.

Lemma 7.2 (Uniform analytic amplitude). *Fix $R > 0$ and $\delta > 0$. If $k \geq \delta n$ and $n - k \geq \delta n$, then, uniformly on $|z| \leq R$,*

$$H_{n,k}(z) = e^{Lz/2} \left[1 + \frac{L^2 z^2 / 12 - Lz/2}{n} + O_{R,\delta}(n^{-2}) \right]. \quad (7.8)$$

The same expansion may be differentiated any fixed number of times, with the corresponding differentiated error bound.

Proof. The unsigned Stirling coefficient is the elementary symmetric function $\begin{bmatrix} n \\ n-d \end{bmatrix} = e_d(1, \dots, n-1)$. Therefore

$$e_d(1, \dots, n-1) \leq \frac{(n(n-1)/2)^d}{d!} \leq \frac{n^{2d}}{2^d d!}.$$

Because $n - d \geq k \geq \delta n$, the absolute value of the d th term in (7.6) is at most $(LR/(2\delta^2))^d / d!$. Its tail for $d > n^{1/10}$ is smaller than every fixed inverse power of n .

For $d \leq n^{1/10}$ we claim the uniform expansions

$$\begin{bmatrix} n \\ n-d \end{bmatrix} = \frac{n^{2d}}{2^d d!} \left[1 - \frac{d(2d+1)}{3n} + O\left(\frac{(1+d)^4}{n^2}\right) \right], \quad (7.9)$$

$$\left(\frac{(n-d)!}{n!} \right)^2 = n^{-2d} \left[1 + \frac{d(d-1)}{n} + O\left(\frac{(1+d)^4}{n^2}\right) \right]. \quad (7.10)$$

For the first, set $S_1 = n(n-1)/2$ and sample d independent indices in $\{1, \dots, n-1\}$ with probabilities $p_j = j/S_1$. The probability of no equal pair is $d!e_d/S_1^d$. The first two inclusion-exclusion bounds for pair coincidences give

$$\mathbb{P}(\text{no coincidences}) = 1 - \binom{d}{2} \sum_j p_j^2 + O\left(d^4 \left(\sum_j p_j^2\right)^2 + d^3 \sum_j p_j^3\right).$$

Here $\sum p_j^2 = 4/(3n) + O(n^{-2})$ and $\sum p_j^3 = O(n^{-2})$. Expanding S_1^d proves (7.9). For the second, expand $-2 \sum_{j=0}^{d-1} \log(1 - j/n)$ with its uniformly bounded Taylor remainder.

Multiplication gives

$$L^d \begin{bmatrix} n \\ n-d \end{bmatrix} \left(\frac{(n-d)!}{n!} \right)^2 = \frac{(L/2)^d}{d!} \left[1 + \frac{d(d-4)}{3n} + O\left(\frac{(1+d)^8}{n^2} \right) \right].$$

The factorial majorant permits summation of the errors and extension of the main terms to all $d \geq 0$. Since $\sum d(d-4)x^d/d! = (x^2 - 3x)e^x$, this proves (7.8). Establishing the same uniform estimate on a slightly larger disk and using Cauchy's derivative estimate proves the differentiated versions. $\square$

Remark 7.3 (The hypothesis δ is not needed). [Added 3 October 2026, batch 85B.] The coefficient of z^d in (7.6) is $\chi_{n,d} L^d$, where $\chi_{n,d} = \begin{bmatrix} n \\ n-d \end{bmatrix} / (n)_d^2$ does not depend on k ; so $H_{n,k}(z)$ is the truncation at degree $n-k$ of the entire polynomial $W_n(Lz)$ of Part II, $W_n(w) = \sum_{d=0}^{n-1} \chi_{n,d} w^d$. Since $f(z)^k$ begins in degree k , the truncation does not change the coefficient in (7.7), which therefore holds with $W_n(Lz)$ in place of $H_{n,k}(z)$; this is manuscript 01's form of the reduction. Manuscript 01 proves the expansion of W_n to every order, uniformly on every compact subset of $\mathbb{C}$ and *with no hypothesis on k* , together with an exact differential recurrence in n (Theorem 14.3 and Lemma 14.2). That is the general form of Lemma 7.2; the conditions $k \geq \delta n$, $n-k \geq \delta n$ above serve only the truncated polynomial, and on compact rays, where both apply, the two statements coincide. The polynomials P_1, P_2 of (7.15) are manuscript 01's p_1, p_2 ; Part II adds P_3 .

7.3 The saddle calculation and its error

The function $r/(1 - e^{-r})$ increases from 1 to ∞ , because its derivative is

$$\frac{1 - (1+r)e^{-r}}{(1 - e^{-r})^2} > 0.$$

Hence the positive saddle is unique. For $\lambda \in K$ it lies in a compact positive interval and $b = r \, d\lambda/dr$ is bounded away from zero.

On the circle $z = re^{i\theta}$, the coefficient integral has phase

$$\log \frac{f(re^{i\theta})}{f(r)} - i\lambda\theta = -\frac{b}{2}\theta^2 - \frac{i\kappa_3}{6}\theta^3 + \frac{\kappa_4}{24}\theta^4 + O_K(\theta^5). \quad (7.11)$$

Every coefficient of f in positive degree is strictly positive. Equality in $|f(re^{i\theta})| \leq f(r)$ can therefore occur only at $\theta = 0$ modulo 2π . Compactness gives a uniform strict gap on every fixed complementary arc. Near zero, the modulus is bounded by $f(r)e^{-c\theta^2}$ for some uniform $c > 0$.

For $g(z) = e^{Lz/2}$,

$$\frac{g(re^{i\theta})}{g(r)} = 1 + iv\theta - \frac{v+v^2}{2}\theta^2 + O_K(\theta^3).$$

Expanding (7.11), multiplying by this amplitude, and setting $\theta = x/\sqrt{k}$ gives the order-1/ k contribution

$$\frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v+v^2}{2b}.$$

These constants follow from the Gaussian moments $\mathbb{E}X^2 = b^{-1}$, $\mathbb{E}X^4 = 3b^{-2}$, and $\mathbb{E}X^6 = 15b^{-3}$ for $X \sim N(0, b^{-1})$. The order-1/ n term of Lemma 7.2, evaluated at the saddle, is $(v^2/3 - v)/n$. Since $n = \lambda k$, this proves the coefficient in (7.4).

For completeness, the error is relative $O_K(k^{-2})$. Expand the analytic phase and amplitude through sufficiently many terms, for example through degree eight, on a shrinking central arc. Odd terms integrate to zero by symmetry, and the Gaussian modulus bound dominates the integrated remainders. The fixed complementary arc is exponentially small; the intervening arc is controlled by the same Gaussian bound. Finally, an analytic amplitude error uniformly $O(n^{-2})$ contributes at most

$$O(n^{-2})f(r)^k r^{-n} \int_{-\pi}^{\pi} \left| \frac{f(re^{i\theta})}{f(r)} \right|^k d\theta = O(n^{-2}k^{-1/2})f(r)^k r^{-n}.$$

Retaining this Gaussian factor proves the required relative error; a bare maximum-modulus estimate would lose an unnecessary factor $\sqrt{k}$. Together with (7.7), this proves the first-order assertion of Theorem 7.1.

[Added 3 October 2026, batch 85B: Theorem 7.1, including its all-orders form (7.17), is also manuscript 01's main theorem on compact rays, with the leading approximation $\mathcal{B}(n, k)$ of (15.1) and the same first coefficient: its $A_1(\rho)$ is $E_1(\lambda)$ of (7.4), term for term, after the renames $\rho \rightarrow \lambda$, $\tau \rightarrow r$, $\mu \rightarrow v$. Its route differs in two places. It integrates the untruncated kernel $W_n(Lz)$ on $|z| = r$ and controls the tails with $|W_n(w)| \leq e^{|w|}$ (Lemma 14.2); on the central arc $|\theta| \leq k^{-2/5}$ it expands two orders beyond the target and bounds the intermediate region by $\exp(-ck^{1/5})$. It also proves the uniqueness of the maximum on the circle directly: equality in the triangle inequality for $\sum_{j \geq 1} r^j e^{ij\theta}/j!$ forces all $e^{ij\theta}$ to agree, and consecutive powers force $e^{i\theta} = 1$.]

7.4 A constructive expansion to every fixed order

Here is a precise algorithm for the higher coefficients, together with the uniform remainder argument. Put $w = Lz$ and $\epsilon = 1/n$. The exact polynomial

$$E_n(w) = \prod_{j=1}^{n-1} \left(1 + \frac{jw}{n^2} \right)$$

has, on every fixed disk, an expansion $E_n(w) \sim e^{w/2} \sum_{p \geq 0} R_p(w) n^{-p}$, with polynomial R_p . Compute them from

$$\log E_n(w) = \sum_{m \geq 1} \frac{(-1)^{m-1} w^m}{m n^{2m}} \sum_{j=1}^{n-1} j^m. \quad (7.12)$$

The inner power sum is a finite polynomial in n ; at order n^{-p} only $m \leq p+1$ is needed. For example,

$$R_0 = 1, \quad R_1 = -\frac{w}{2} - \frac{w^2}{6}, \quad R_2 = \frac{w^2}{4} + \frac{w^3}{12} + \frac{1}{2} \left(\frac{w}{2} + \frac{w^2}{6} \right)^2.$$

Define polynomials $Q_q(d)$ by the formal identity

$$\exp \left(2 \sum_{m \geq 1} \frac{\epsilon^m}{m} \sum_{j=0}^{d-1} j^m \right) = \sum_{q \geq 0} Q_q(d) \epsilon^q. \quad (7.13)$$

Thus $Q_0 = 1$, $Q_1 = d(d-1)$, and

$$Q_2 = \frac{d(d-1)(2d-1)}{6} + \frac{d^2(d-1)^2}{2}.$$

With $D_w = w d/dw$, put

$$P_j(w) = e^{-w/2} \sum_{p+q=j} Q_q(D_w) \left[e^{w/2} R_p(w) \right]. \quad (7.14)$$

These are polynomials, and

$$H_{n,k}(z) \sim e^{Lz/2} \sum_{j \geq 0} \frac{P_j(Lz)}{n^j}, \quad P_0 = 1, \quad P_1(w) = \frac{w^2}{12} - \frac{w}{2}, \quad P_2(w) = \frac{w^4}{288} - \frac{w^2}{12}. \quad (7.15)$$

To justify this algorithm to any fixed order p , set $F_d(x) = \prod_{j=0}^{d-1} (1-jx)^{-2}$. For $d \leq n-k \leq (1-\delta)n$ and $0 \leq x \leq 1/n$, every $1-jx \geq \delta$. The m th derivative of its logarithm is

$$2(m-1)! \sum_{j < d} \frac{j^m}{(1-jx)^m} = O_{m,\delta}((1+d)^{m+1}).$$

The finite product rule for derivatives of an exponential then gives

$$|F_d^{(p+1)}(x)| \leq C_{p,\delta} \delta^{-2d} (1+d)^{2p+2}.$$

Taylor's theorem bounds the remainder after the p th term by $C_{p,\delta} n^{-p-1} \delta^{-2d} (1+d)^{2p+2}$. Multiplication by the exact coefficient $e_d(1, \dots, n-1)/n^{2d} \leq 1/(2^d d!)$ makes this error summable on every fixed w -disk. Multiplying its d th coefficient by $Q_q(d)$ is exactly the operation $Q_q(D_w)$. Extending the sum beyond $n-k \geq \delta n$ adds only a factorially small tail. Independently, expanding (7.12) and its derivatives on a slightly larger disk has a uniform Taylor remainder. Combining the two steps proves (7.15), with error $O(n^{-p-1})$ after any fixed order p .

The remaining Gaussian integrations are also explicit. Define the linear functional on polynomials in y by

$$\mathcal{G}_b(y^{2m}) = (-1)^m (2m-1)!! b^{-m}, \quad \mathcal{G}_b(y^{2m+1}) = 0,$$

with $(-1)!! = 1$. It is integration after the substitution $y = iX$, $X \sim N(0, b^{-1})$. Introduce a formal variable τ and set

$$E_j(\lambda) = \mathcal{G}_b \left([\tau^{2j}] \exp \left\{ \sum_{m \geq 3} \frac{\kappa_m y^m \tau^{m-2}}{m!} + v(e^{\tau y} - 1) \right\} \times \sum_{p \geq 0} \frac{\tau^{2p}}{\lambda^p} P_p(Lre^{\tau y}) \right). \quad (7.16)$$

At each order this is a finite polynomial calculation: it uses $p \leq j$ and $m \leq 2j+2$. Odd powers vanish after applying $\mathcal{G}_b$. It gives $E_0 = 1$ and the E_1 in (7.4). Therefore for every fixed $J \geq 0$,

$$T(n, k) = \frac{n! f(r)^k e^v}{2L^{n+1} r^n \sqrt{2\pi k b}} \left[\sum_{j=0}^J \frac{E_j(\lambda)}{k^j} + O_{K,J}(k^{-J-1}) \right]. \quad (7.17)$$

The same arc estimates as above justify termwise finite integration. This is an asymptotic expansion to each finite order; convergence of the infinite displayed series is not asserted.

[Added 3 October 2026, batch 85B: manuscript 01 gives the same finite prescription with its expansion parameter called ε instead of τ : its $H_M(\varepsilon, y)$ is the expression inside $\mathcal{G}_b$ in (7.16), truncated at $m \leq 2J+2$ and $p \leq J$, because $P_p(Lre^{\tau y})$ is its $p_p(2\mu e^{\varepsilon y})$ with $2\mu = Lr$. It stresses the order of operations: first expand and truncate in τ , then apply $\mathcal{G}_b$ to each polynomial coefficient; no Gaussian expectation of an untruncated exponential of a high-degree polynomial is intended, and such an integral need not exist. It computes the kernel polynomials by a different finite route, (14.11)–(14.12), and adds a polynomial recursion for the κ_j ((15.2)); both are in Part II.]

7.5 The A261784 diagonal

Put $n = 2N$, $k = N$, and

$$r = 2 + W_0(-2e^{-2}) = 1.5936242600400400923230 \dots \quad (7.18)$$

The principal real Lambert branch is necessary; the other real branch gives the excluded boundary value $r = 0$. Equation (7.1) becomes $r/(1 - e^{-r}) = 2$. Define

$$d_* = \frac{4(e^r - 1)}{L^2 r^2} = \frac{4}{L^2 r(2 - r)} = 12.8556891736661213193266 \dots, \quad (7.19)$$

$$c_* = \frac{2^{r/2}}{4\pi L \sqrt{r - 1}} = 0.2588649231114655502517752 \dots \quad (7.20)$$

Corollary 7.4 (Exact amplitude and correction for A261784). *The sequence $a_N = T(2N, N)$ satisfies*

$$a_N = c_* d_*^N \frac{(N!)^2}{N} \left(1 + \frac{\beta}{N} + O(N^{-2}) \right), \quad (7.21)$$

where, with $v = Lr/2$,

$$\begin{aligned} b &= 2(r - 1), \quad \kappa_3 = 2(r^2 - 3r + 3), \quad \kappa_4 = 2(r^3 - 8r^2 + 19r - 13), \\ \beta &= \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v + v^2}{2b} + \frac{v^2/3 - v}{2} - \frac{1}{8} \\ &= -0.5218067015564826638891441 \dots \end{aligned}$$

Proof. Specialize Theorem 7.1 at $\lambda = 2$ and use

$$\frac{(2N)!}{(N!)^2} = \frac{4^N}{\sqrt{\pi N}} \left(1 - \frac{1}{8N} + O(N^{-2}) \right).$$

The saddle prefactor has $b = 2(r - 1)$. Collecting the two square roots gives (7.20), and the central-binomial correction accounts for the final $-1/8$ in β . $\square$

The OEIS entry already gives the leading equivalent in (7.21), with d_* in Lambert- W form and c_* numerically [3]. The contribution here is its derivation from the positive transform, the exact expression (7.20), the explicit correction, and its place in the uniform all-orders theorem.

[Added 3 October 2026, batch 85B: manuscript 01 proves the same corollary to every fixed order, with the same r , d_* , c_* and first correction; written out with $\kappa_3 = 2q$, $\kappa_4 = 2s$, $q = r^2 - 3r + 3$, $s = r^3 - 8r^2 + 19r - 13$, its closed form is $\beta_1 = \frac{s}{16(r-1)^2} - \frac{5q^2}{48(r-1)^3} - \frac{v^2+v}{4(r-1)} + \frac{qv}{4(r-1)^2} + \frac{v^2}{6} - \frac{v}{2} - \frac{1}{8}$, which is the display above with $b = 2(r - 1)$ substituted. Both manuscripts note that the expression for d_* agrees algebraically with the one posted in A261784 and that c_* reproduces its decimal prefactor. The two computations of r , d_* , c_* , β_1 and β_2 were made by independent programs and agree in every printed digit. Manuscript 01 adds the third correction β_3 , the recipe for every further one, and fifty-digit values (Section 17).]

7.6 Column counts under the dense row condition

There is a further consequence: enforcing every row to be nonzero in this regime still leaves a universal leading Gaussian column law.

Corollary 7.5. *For a uniformly chosen matrix counted by $T(n, k)$, with $n/k \in K \Subset (1, \infty)$, its number of columns satisfies*

$$\frac{C - n/(2L)}{\sqrt{n(1-L)/(4L^2)}} \xrightarrow{d} N(0, 1). \quad (7.22)$$

Uniformly in this regime,

$$\mathbb{E}C = \frac{n+1}{2L} - \frac{1}{2} - \frac{r}{4} + O_K(k^{-1}), \quad (7.23)$$

$$\text{Var}(C) = \frac{(n+1)(1-L)}{4L^2} - \frac{1}{4} + \frac{r}{8} + O_K(k^{-1}). \quad (7.24)$$

Proof. The weighted version of (2.4) replaces F_q by $F_q(u) = \sum_c c! \{c\}^q u^c$, whose exponential generating function is $1/(1+u-ue^z)$. Its dominant pole is $L_u = \log(1+1/u)$ with residue $-1/(1+u)$; the other poles are $L_u + 2\pi i j$. The same positive-sum and saddle proof works uniformly for real u in a compact interval around 1. It gives

$$T(n, k; u) = \frac{n! f(r)^k e^{L_u r/2}}{(1+u) L_u^{n+1} r^n \sqrt{2\pi k b}} (1 + O_K(k^{-1})). \quad (7.25)$$

In the ratio of (7.25) to its value at $u = 1$, put $u = e^{t/\sqrt{n}}$. Direct Taylor expansion of the elementary function $\log(L/L_{e^v})$ has first derivative $1/(2L)$ and second derivative $(1-L)/(4L^2)$ at zero. Hence

$$\log \mathbb{E} \exp \left(\frac{t(C - n/(2L))}{\sqrt{n}} \right) \rightarrow \frac{t^2(1-L)}{8L^2}.$$

This proves the Gaussian limit by real moment generating functions; no differentiation of an unspecified asymptotic error is used.

For the sharper moment statements, use an exact auxiliary mixture. Let Q have probabilities proportional to $\binom{n}{q} \{q\} F_q$, $k \leq q \leq n$, and put $D = n - Q$. Conditional on $Q = q$, the distribution of C is the block-count distribution J_q in a uniform ordered set partition of q elements, because its probability generating polynomial is $F_q(u)/F_q$. This specifies an exact distributional mixture; no canonical matrix statistic is being claimed for D .

Marking D with a fresh variable ξ replaces $H_{n,k}(z)$ by $H_{n,k}(\xi z)$. Applying the coefficient proof separately to $zH'_{n,k}(z)$ and $z^2H''_{n,k}(z)$ gives

$$\mathbb{E}D = \frac{Lr}{2} + O_K(k^{-1}), \quad \text{Var}(D) = \frac{Lr}{2} + O_K(k^{-1}).$$

Here the differentiated analytic kernel estimates follow from Lemma 7.2; the exponentially small Fubini replacement error is also valid for each of these nonnegative weighted sums. The exact identities

$$F'_q(1) = \frac{F_{q+1} - F_q}{2}, \quad F''_q(1) = \frac{F_{q+2} - 5F_{q+1} + 2F_q}{4}$$

follow from the Stirling recurrence or direct differentiation of the generating function. Using (7.5), they yield

$$\mathbb{E}J_q = \frac{q+1}{2L} - \frac{1}{2} + O(q\vartheta^q), \quad \text{Var}(J_q) = \frac{(q+1)(1-L)}{4L^2} - \frac{1}{4} + O(q^2\vartheta^q).$$

Since $Q \geq k$, the exponential errors stay negligible after mixing. The laws of total expectation and total variance now give (7.23)–(7.24). $\square$

[Added 3 October 2026, batch 85B: the auxiliary variable $D = n - Q$ of this proof is manuscript 01's *Stirling defect* $\Delta_{n,k}$ (with $Q = Q_{n,k}$). Manuscript 01 proves more than its two moments: $\Delta_{n,k}$ converges in law to $\text{Poisson}(Lr/2)$, and its probability generating function is $e^{v(\xi-1)}(1 + O(k^{-1}))$ uniformly for complex ξ in compact sets, with an expansion to every fixed order (Theorem 16.2). The two displayed moments follow from it. Both manuscripts state that no canonical matrix statistic is claimed for this variable.]

The compact restriction $K \subseteq (1, \infty)$ is essential in all these statements. At $k = n$ the Stirling kernel collapses and $T(n, n) = \mathbf{F}_n$ exactly. At the coverage window $n/k \sim \log k$ the saddle leaves compact sets. Neither boundary is covered by the uniformity asserted in Theorem 7.1.

8 Prescribing the number of columns changes the threshold

The preceding models allow the column count to fluctuate. A different coverage scale appears when it is prescribed to be much smaller than the number of rows. The exact stars-and-bars identities underlying this model are classical [4, Proposition 28]; the following limit follows from direct uniform estimates on them.

Theorem 8.1 (Fixed-column coverage). *Let $k \rightarrow \infty$, let $m = m(k) \geq 1$ be integers with $m = o(k)$, and let $n = n(k) \geq 1$. Choose uniformly a $k \times m$ matrix of nonnegative integers of total sum n , conditioned on no zero column. Let M count its zero rows and put*

$$h = \frac{n}{k}, \quad \theta = m \log \left(1 + \frac{n}{km} \right). \quad (8.1)$$

If $\theta - \log k \rightarrow s \in \mathbb{R}$, then

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \mathbb{P}(M = 0) \rightarrow \exp(-e^{-s}).$$

Thus an exact parametrization of the critical mass, up to integer rounding, is

$$n = km \left(\exp \left(\frac{\log k + s}{m} \right) - 1 \right). \quad (8.2)$$

Proof. First allow zero columns, and call the zero-row count M_0 . There are $\binom{n+km-1}{n}$ matrices. If j designated rows are zero, deleting them and comparing these binomial coefficients gives, with $N = km$,

$$\mathbb{E}[(M_0)_j] = (k)_j \prod_{a=1}^{jm} \frac{N - a}{N + n - a}. \quad (8.3)$$

Fix j and take $k \geq 2j$. For $f(a) = \log(N - a) - \log(N + n - a)$ one has $f(0) = -\log(1 + n/N)$ and

$$f'(a) = -\frac{n}{(N - a)(N + n - a)}.$$

The mean value theorem and summation imply

$$\sum_{a=1}^{jm} (f(a) - f(0)) = O_j \left(\frac{nm^2}{N(N + n)} \right) = O_j \left(\frac{mh}{k(m + h)} \right).$$

Also $\log(k)_j = j \log k + O_j(k^{-1})$. Therefore

$$\log \mathbb{E}[(M_0)_j] = j(\log k - \theta) + O_j \left(\frac{1 + \min(m, h)}{k} \right).$$

For $x \geq 0$, $\log(1+x) \geq (\log 2) \min(1, x)$, so $\theta \geq (\log 2) \min(m, h)$. At the assumed critical scale, the error is $O_j((\log k)/k)$. The factorial moments tend to e^{-js} ; the finite Taylor/Bonferroni argument in Lemma 6.1 proves the Poisson limit.

It remains to justify conditioning. If $m = 1$, the sole column cannot be zero because $n > 0$. If $m \geq 2$, the probability that one designated column is zero is at most

$$\prod_{a=1}^k \frac{N-a}{N+n-a} \leq \left(\frac{N}{N+n} \right)^k = \exp(-k\theta/m).$$

The union bound gives probability at most $m \exp(-k\theta/m)$ for any zero column. Its logarithm tends to $-\infty$, since $m = o(k)$ and $\theta = \log k + O(1)$. Conditioning on an event whose probability tends to one changes the law of any statistic in total variation by a quantity tending to zero. Thus M and M_0 have the same limit. Rounding n in (8.2) changes θ by at most $1/k$. $\square$

Corollary 8.2 (A shifted window). *Suppose $m/(\log k)^2 \rightarrow \gamma \in (0, \infty)$ and $n/k - \log k \rightarrow s$. In the preceding fixed-column model,*

$$M \xrightarrow{d} \text{Poisson}\left(e^{-s+1/(2\gamma)}\right), \quad \mathbb{P}(M=0) \rightarrow \exp\left(-e^{-s+1/(2\gamma)}\right). \quad (8.4)$$

Proof. Expanding (8.1) gives

$$\theta = h - \frac{h^2}{2m} + O(h^3/m^2) = \log k + s - \frac{1}{2\gamma} + o(1).$$

Apply Theorem 8.1 with this shifted constant. $\square$

Other regimes follow from the exact parametrization. If $(\log k)^2 = o(m)$ and $m = o(k)$, the usual critical mass $k(\log k + s) + o(k)$ returns. If $m \sim \gamma \log k$, the leading critical mass is $\gamma(e^{1/\gamma} - 1)k \log k$; the smaller order- k centering depends on the actual integer sequence $m(k)$, so the exact formula should be retained for that precision. For a fixed positive integer m , the critical mass is

$$n = m e^{s/m} k^{1+1/m} - mk + O(1).$$

In particular, one column requires mass of order k^2 for coverage.

This is a theorem about a prescribed column count. It is not a theorem about a column weight $u = u_k \rightarrow 0$; relating those two ensembles is a separate research problem.

9 Exact checks and numerical diagnostics

The accompanying programs reproduce the integer checks and numerical tables independently of the typesetting. Integer recurrences, polynomial identities, and determinants are evaluated exactly. The subsequent asymptotic comparisons use 120 decimal digits of working precision. Their role is to detect implementation errors and illustrate the proved estimates; the numerical comparisons do not constitute proofs.

[Added 3 October 2026, batch 85B: this section uses manuscript 06's delivery names. In the report, `code/verify.py` is `code/06-hankel-verify.py`, `code/make_figures.py` is `code/06-hankel-make_figures.py`, each `data/X` is `data/06-hankel-X`, `code/requirements.txt` is `data/06-hankel-requirements.txt`, the figure is `figures/06-hankel-coupon_window.pdf`, and `README.txt` is replaced by the report's `README.md`, which gives rerun instructions that work with these names. Manuscript 01's checks are described in Section 19.]

9.1 Independent exact calculations

The program `code/verify.py` constructs $A(n, k)$ from the nonzero-column composition recurrence (2.2) and obtains $T(n, k)$ by binomial inversion (2.3). Separately, it constructs both Stirling triangles and the Fubini numbers by their integer recurrences, and evaluates the positive transform (2.4). These two routes agree for all $0 \leq n \leq 30$, $0 \leq k \leq 10$: 341 comparisons for each of A and T . All nine supplied rows $n = 0, \dots, 8$ of A261781 also agree exactly [2].

The recurrence checks cover $k = 1, \dots, 10$, with coefficient arrays computed through $n = 85$. They verify coprimality of the exact numerator and denominator, the nonzero impulse at degree $D = k(k+1)/2$, and the homogeneous recurrence beginning at $D+1$, as in Theorem 2.2. The determinant checks verify Theorem 3.1 for $k = 1, \dots, 5$ and shifts 1, 2, 3; the next-size tail determinants vanish for $k \leq 4$. They also check the initial size- $(D+1)$ identity (3.10) at $u = 1$ for $k \leq 4$. Five further cases are exact polynomial identities in the column weight u , including

$$H_{3,1}(u) = 27u^{14}(1+u)^8(1+2u)^2.$$

These finite checks agree with the general algebraic proofs; they do not replace the uniform arguments in those proofs.

9.2 The coverage window

Write $p_{n,k} = T(n, k)/A(n, k)$ and, using the actual integer value of n , set

$$h = \frac{n}{k}, \quad \eta = ke^{-h}, \quad p_0 = e^{-\eta}, \quad p_1 = e^{-\eta} \left(1 + \frac{(1-L)h\eta - (1+h)\eta^2}{2k} \right).$$

Thus p_0 is the moving-mean Poisson approximation and p_1 includes the first correction in Theorem 5.1. Table 3 gives the signed relative errors $\delta_0 = p_{n,k}/p_0 - 1$ and $\delta_1 = p_{n,k}/p_1 - 1$. Here n is the nearest integer to $k \log k$, so the target window coordinate is zero but its actual value $s = n/k - \log k$ is retained. This avoids assigning integer-rounding effects to the asymptotic error.

k	n	Actual s	$p_{n,k}$	δ_0	δ_1
20	60	+0.004268	0.339714114626	-8.04855×10^{-2}	-4.60223×10^{-3}
50	196	+0.007977	0.356935471459	-3.74272×10^{-2}	-9.74476×10^{-4}
100	461	+0.004830	0.361879493451	-2.10377×10^{-2}	-3.02538×10^{-4}
500	3107	-0.000608	0.365694896898	-5.33335×10^{-3}	-1.86151×10^{-5}
1000	6908	+0.000245	0.366903156844	-2.89783×10^{-3}	-5.38847×10^{-6}

Table 3: Coverage diagnostics at the rounded central window. The displayed probabilities and errors are rounded from 120-digit evaluations; both approximations use the actual n .

The probability calculation uses explicit error accounting. Put $E_n = B\rho^{n-1}$, with the constants of Theorem 4.1. For $1 \leq j < k$ define

$$\tilde{m}_j = \binom{k}{j} \frac{c_{k-j} d_{k-j}^n}{c_k d_k^n}, \quad m_j = \mathbb{E} \binom{M}{j}.$$

Whenever $E_n < 1$, the dominant-pole estimate gives

$$|m_j - \tilde{m}_j| \leq \frac{2E_n}{1 - E_n} \tilde{m}_j.$$

Odd and even Bonferroni sums, enlarged by the summed error on their terms, therefore give analytic lower and upper bounds. The computation stops when the last retained moment is below 10^{-70} , or at $j = k$, where the finite inclusion–exclusion sum is complete. All 15 tabulated cases at target $s = -1, 0, 1$ are supplied in the data files. Five cases with $n \leq 200$ were also compared against independently computed exact integer ratios. Thirty additional evaluations check the uniform pole bound against exact unrestricted coefficients.

The truncation and pole-replacement errors are thus bounded analytically. The decimal endpoints themselves use high-precision floating arithmetic and have *not* been certified by directed rounding. The stated numerical precision is a diagnostic setting, not an interval-arithmetic certificate.

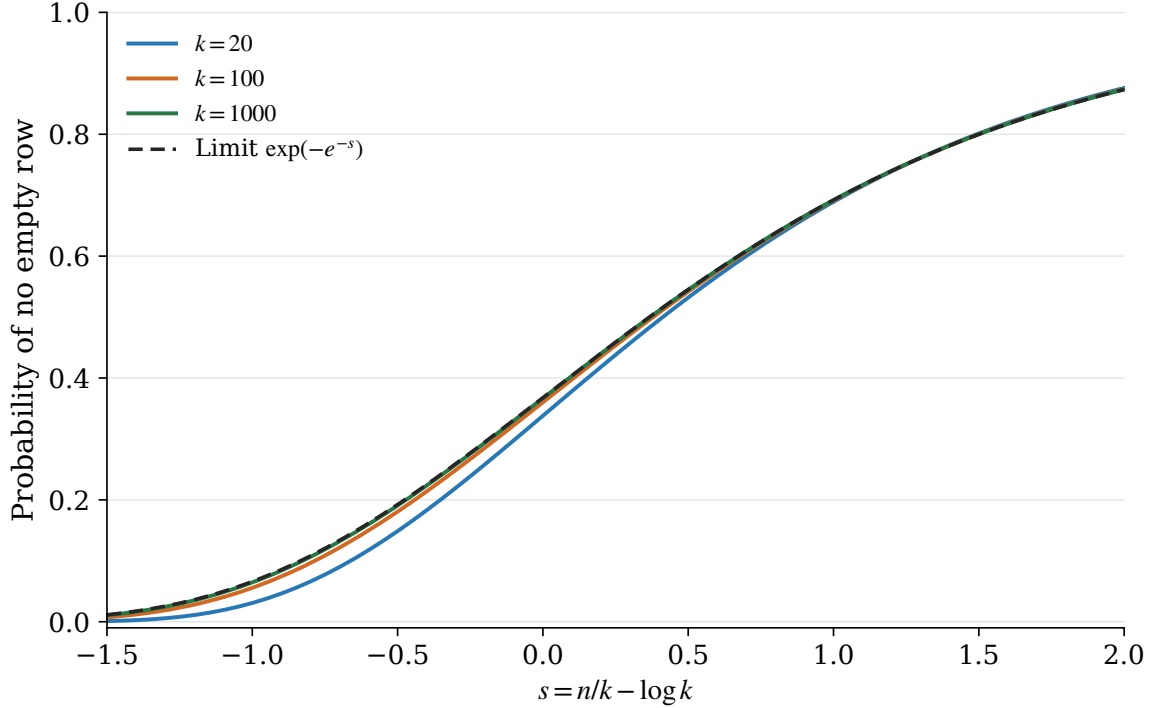


Figure 1: The empty-row transition. Finite-height curves connect evaluations at integer masses for $k = 20, 100, 1000$; the dashed curve is $\exp(-e^{-s})$. The horizontal coordinate always uses the actual $s = n/k - \log k$. The probabilities are evaluated by the bounded Bonferroni calculation described above.

9.3 The dense diagonal and two correction coefficients

For A261784 define the normalized exact count

$$\mathcal{R}_N = \frac{T(2N, N)}{c_* d_*^N (N!)^2 / N},$$

with c_*, d_* from (7.20) and (7.19). The all-orders expansion (7.17), specialized at $\lambda = 2$, gives

$$\mathcal{R}_N = 1 + \frac{\beta_1}{N} + \frac{\beta_2}{N^2} + O(N^{-3}), \quad \beta_1 = \beta.$$

There is an exact coefficient rule for the second constant:

$$\beta_1 = E_1(2) - \frac{1}{8}, \quad \beta_2 = E_2(2) - \frac{E_1(2)}{8} + \frac{1}{128}. \quad (9.1)$$

Here E_1, E_2 are obtained from the finite Gaussian-moment rule (7.16); the last two terms arise from the central-binomial expansion. Evaluating that rule, independently of the diagonal integer data, gives

$$\begin{aligned}\beta_1 &= -0.521806701556482663889144103069\dots, \\ \beta_2 &= 0.128288828088705778459796704234\dots\end{aligned}$$

Thus β_2 is a derived coefficient rather than a fitted constant. The finite arithmetic expression used for its evaluation is included in `code/verify.py`.

To display the convergence, define

$$\begin{aligned}U_N &= N(\mathcal{R}_N - 1), & V_N &= N^2 \left(\mathcal{R}_N - 1 - \frac{\beta_1}{N} \right), \\ W_N &= N^3 \left(\mathcal{R}_N - 1 - \frac{\beta_1}{N} - \frac{\beta_2}{N^2} \right).\end{aligned}$$

Theorem 7.1 and its all-orders refinement imply $U_N \rightarrow \beta_1$, $V_N \rightarrow \beta_2$, and boundedness of W_N . The numerator $T(2N, N)$ in every row of Table 4 is computed as an exact integer from (2.4); only the normalization uses floating arithmetic.

N	U_N	V_N	W_N
5	-0.49620219	0.12802256	-0.00133133
10	-0.50899740	0.12809304	-0.00195793
20	-0.51539780	0.12817802	-0.00221621
30	-0.51753296	0.12821232	-0.00229519
50	-0.51924187	0.12824171	-0.00235602
75	-0.52009661	0.12825702	-0.00238568
100	-0.52052405	0.12826482	-0.00240032

Table 4: Dense-diagonal diagnostics from exact integer counts. The first two columns approach the independently derived correction coefficients; the final column records the next scaled remainder.

At $N = 100$, the signed relative error after including β_1/N is 1.28938×10^{-5} ; after also including β_2/N^2 it is -2.41288×10^{-9} . These are finite numerical comparisons, not uniform error constants for the asymptotic theorem.

[Added 3 October 2026, batch 85B: manuscript 01’s Table 7 reports the same comparisons with the opposite sign convention, approximation divided by exact count minus one; at $N = 100$ it lists -1.289×10^{-5} and $+2.413 \times 10^{-9}$, the same two numbers to first order. Its exact counts a_N , $N \leq 200$, come from its own program.]

9.4 Reproduction and data

The package includes the complete L^AT_EX source, compiled article, both Python programs, their pinned dependencies, the figure in PDF and PNG form, and CSV/JSON outputs. From the package root, `python3 code/verify.py` regenerates the checks and tables; `python3 code/make_figures.py` regenerates the plot. The `README.txt` gives the dependency and L^AT_EX build commands. The programs use no network access after their dependencies are installed. The supplied run records report Python 3.12.14, SymPy 1.14.0, mpmath 1.3.0, and matplotlib 3.10.8.

[Added 3 October 2026, batch 85B: in this report the compiled article is a build of the merged text, not manuscript 06’s delivered PDF, and the names are those listed at the start of this section.]

Both programs write unprefixed files under their output root, which defaults to the directory above `code/`, and `06-hankel-make_figures.py` imports its sibling as `verify`; they must therefore be rerun on a copy with the delivered names restored, as the report README describes, never in the report directory. At the intake's rerun on a copy (Python 3.13.5) `verify.py` passed in about 5 s of Python time and the figure program took about 74 s, against the 0.3 s and 2.3 s of the delivered README; the six CSV files were byte-identical, five JSON files equal apart from line endings, and the PNG figure pixel-identical.]

10 Further research: precise questions and useful next steps

The following are proposed research problems motivated by the results above. They are not assertions that every formulation is open throughout the literature. The source audit was narrower than that. Each question identifies a statement not proved in this article and a concrete way in which its resolution would extend the present results.

Research question 1 (A uniform approximation across the triangle). Can one obtain relative asymptotics for $T(n, k)$ that remain valid from the near-diagonal boundary $n - k = o(k)$ through the dense rays and into the coverage window $n/k = \log k + O(1)$?

There are two distinct boundaries to resolve. As $n/k \downarrow 1$, the saddle r tends to zero, the variance parameter b tends to zero, and the compact-set saddle proof no longer applies. At the endpoint, $T(n, n) = F_n$ exactly. As n/k grows, the saddle escapes to infinity and the kernel estimates must hold on expanding disks. A useful first deliverable is a uniform formula for $n = k + \ell$ with ℓ growing through the range where a fixed-defect expansion ceases to be accurate. A second is a proved overlap region in which the positive-transform saddle estimate and the empty-row probability estimate agree, including their relative errors. Such a result would turn the separate formulas in this article into a controlled asymptotic description of a much larger part of A261781.

Research question 2 (Wider windows and higher coverage corrections). For which explicit ranges of $s_k = n/k - \log k$ can the Poisson–Charlier expansion of Section 5 be extended, and can all its correction polynomials be generated with a uniform summable remainder?

The theorem here keeps $\eta = ke^{-n/k}$ in a fixed compact subset of $(0, \infty)$. If η grows, the factorial-moment sums are concentrated at growing indices, so the same fixed-disk argument does not directly give a useful relative estimate for the small probability $M = 0$. If η tends to zero, a different error scale is appropriate. Expanding the exact ratios $c_{k-j}d_{k-j}^n/(c_k d_k^n)$ suggests a constructive series in Poisson–Charlier polynomials, but the admissible joint growth of j , η , and h must be proved. A satisfactory result would state both an explicit range and a remainder that remains smaller than the probabilities or distances being approximated.

[Added 3 October 2026, batch 85B: for the single event $M = 0$ and bounded s , the second half of this question is answered by manuscript 01: Theorem 13.1 gives every correction polynomial of $\mathbb{P}(M = 0)$ by a finite algorithm, with a uniform remainder $O((1+h)^{J+1}/k^{J+1})$ after order J . The question stays open for the whole law of M (the generating function (5.3) and the probability-mass expansion beyond the first correction) and for every window wider than bounded s .]

Research question 3 (Joint local limits and residual dependence). Can the Gaussian–Poisson convergence in Section 6 be strengthened to a joint local limit theorem with an explicit first correction?

For example, one can ask for a uniform approximation to $\mathbb{P}(C = c, M = m)$ for integer c in a growing central range and m in a specified range, and for the corresponding law conditioned on $M = 0$. The present moment-generating-function proof establishes distributional independence after normalization, but it does not control the Fourier integral away from the origin. A local theorem requires both this off-axis control and a lattice analysis. The first mixed correction would quantify the small dependence that survives before taking the limit, and would refine the order-one column-moment formulas on dense rays. This is a natural setting in which to compare joint Edgeworth and Poisson–Charlier expansions.

Research question 4 (Small column weights and equivalence of ensembles). How does the coverage transition change when the column weight $u = u_k$ tends to zero, and when does conditioning its weighted model on $C = m$ preserve the corresponding asymptotic laws?

For any fixed m , conditioning on $C = m$ removes the factor u^m , so the resulting distribution is exactly the prescribed-column model. The analytic difficulty is making that conditioning asymptotically uniform when its probability may be small. The weighted pole estimate in Section 4 is uniform on compact positive weight intervals; it does not supply a uniform spectral gap as $u \downarrow 0$. The explicit critical parameter $m \log(1 + n/(km))$ and its shift at $m \asymp (\log k)^2$ provide testable targets for a joint saddle analysis in z and u . Extending the prescribed-column theorem from $m = o(k)$ to comparable or larger column counts is another concrete part of this problem.

Research question 5 (A combinatorial explanation of determinant positivity). Is there a direct combinatorial model for the coefficients of $H_{k,s}(u)$, or a sign-reversing involution proving the product in Theorem 3.1?

The entries of the determinant count matrices, but the determinant itself is a signed sum of products of such counts. Its coefficientwise positivity is presently explained by spectral factorization. A direct model would explain the factors $\binom{k}{j}^j$ and the powers of $(1 + u)^m - u^m$, and might expose refinements that the one-variable spectrum does not reveal. The negative smaller minor exhibited in Section 3 rules out a blanket appeal to total positivity. One should seek a model specific to these maximal minors, or a precise classification of which other minors have a definite sign.

Research question 6 (Recurrences in finite characteristic). What is the minimal eventual recurrence of the column $T(n, k)$ over $\mathbb{F}_p$, and how does it depend on k and the prime p ?

Over characteristic zero every pole survives and all root sets are disjoint. Reduction modulo a prime can merge roots, make multiplicities inseparable, or cancel numerator and denominator factors. The determinant product isolates the possible exceptional primes through binomial coefficients, powers of 2, and factors $2^m - 1$. It therefore supplies an arithmetic guide, but does not by itself classify the remaining rank or the reduced rational function at each exceptional prime. An exact classification could yield new divisibility and periodicity statements for the triangle, with fast recurrence-based proofs suitable for individual OEIS entries.

Research question 7 (The limiting distribution of determinant zeros). After removing the factors at $u = 0$ and $u = -1$, what is the limiting zero-counting measure of $H_{k,s}(u)$ as $k \rightarrow \infty$?

Every remaining zero has real part $-1/2$ and is the image of a root of unity under $\zeta \mapsto 1/(\zeta - 1)$. Its multiplicity is controlled by the weighted gcd sum over pairs $1 \leq i < j \leq k$. Determining the normalized measure therefore reduces to a specific arithmetic counting problem, rather than an unknown root-location problem. Useful targets include an explicit density along the vertical line, tail estimates for zeros with large imaginary part, and the contribution of roots of small order. The

answer should specify whether the zero measure is normalized by the full degree or by the degree remaining after the two exceptional factors are removed.

Research question 8 (Which column models have the same coverage law?). Which replacements for the unrestricted column generating function $(1 - z)^{-k} - 1$ retain the window $n = k(\log k + s)$, and which alter its first correction or its scale?

Natural examples impose bounded entries, bounded column sums, or a weight depending on column size. The row symmetry needed for exact factorial moments may survive even when the root-of-unity spectrum does not. One can aim for sufficient analytic conditions on the dominant pole and its row-deletion ratios that imply a Poisson law, then identify which derivatives determine the first correction. A theorem formulated this way would separate properties special to matrix compositions from properties shared by a wider class of symmetric occupancy models.

10.1 A practical order of attack

For a continuation in ProveIt, the most direct algebraic project is to formalize the reduced denominator and its no-cancellation proof, followed by the resultant and maximal Hankel determinant identities. These statements are finite algebra and have explicit base conventions. They provide a manageable route to a proof-assistant-checked contribution without first formalizing saddle-point analysis.

[Added 3 October 2026, batch 85B: manuscript 01 names the same first target (Section 20). Nothing of either manuscript is formalized; the only formal neighbour is the Lean Fubini generating function cited in Section 1, and this paragraph is a proposal, not a plan of the repository.]

For analytic work, the highest-value next targets are the near-diagonal boundary and a proved overlap with the coverage regime. The positive Stirling transform avoids cancellation and gives an explicit starting point in both cases. The small-weight problem is a second substantial direction, because the prescribed-column result already identifies a change of scale that a uniform two-variable theory must explain. The computation scripts in the accompanying package can supply regression examples for these projects, but every extension needs a new proof of its claimed uniformity and error bounds.

Part II

Finite spectra, uniform asymptotics, and two Poisson limits

11 Provenance, scope and notation of Part II

[Added 3 October 2026, batch 85B.] Part II was added in the write of batch 85B. It is built from manuscript 01 of batch 85 and prints everything that manuscript adds to Part I. Where the two manuscripts prove the same statement, the statement is printed once, in Part I, with a bracketed note naming manuscript 01's version; Table 5 lists those second routes. Everything in Part II is written in Part I's letters (Table 2).

Standing assumption: $u = 1$. Manuscript 01 never weights columns. *Every statement of Part II is at column weight $u = 1$:* its $A(n, k)$, $T(n, k)$, c_k , d_k , $F_k(z; 1)$ and M are the unweighted objects

of Part I, and its generating variable ξ in Section 16 marks the Stirling defect, not columns. No weighted analogue of a Part II result is claimed. Several of its proofs use $u = 1$ explicitly: the residue $-a_k/(2k)$, the bound $a_k \geq \frac{1}{2}$, the Fubini pole at $L = \log 2$ and the constants of Section 13 are the $u = 1$ values of Part I's $-a_{k,u}/((1+u)k)$, $a_{k,u} \geq b$, L_u and $\mu_k(u)$, $d_k(u)$.

11.1 The manuscript

Manuscript 01 of batch 85 is the archive `OEIS_Matrix_Compositions_Research.zip` (wrapper directory `matrix_compositions_research/`, 26 files), which arrived in commit `317c1ce2e` and was placed in this report by commit `fa2f3e419`. Its article is titled *Matrix compositions: finite spectra, uniform asymptotics, and two Poisson limits*, subtitled *Proofs for OEIS A261781 and all-orders refinements for A261784* (22 pages, dated 3 October 2026). Its title page and PDF metadata read “Research report prepared for Vladimir Reshetnikov”; it names no author and no tool. Its pin is the repository tree revision `d68b65ea064084fb121df9bb431b21d086f7be81` (2026-10-03, 17:08), at which it read the series-and-transseries overview `Analysis/Transseries/docs/series-and-transseries/README.md` (blob `7bdc83aeb`) as motivation for its inversion section; its indexed searches for A261781 and A261784 returned nothing and exposed a different revision, `ce37e13f4` [9].

It was written without sight of manuscript 06, the source of Part I, which arrived thirteen minutes later in commit `345284ac8`. The two texts share 0.7% (01 in 06) and 0.5% (06 in 01) of their word 8-grams, almost all of them bibliography entries. Every number they share agrees: the constants r , b , v , d_* , c_* , β_1 , β_2 in about seventy printed digits, computed by two independent programs, and the coverage probabilities they both tabulate.

Status of manuscript 01, as delivered. It is a proof-bearing research report, not a Lean-certified or independently refereed publication. The two OEIS conjecture labels were present in the entry it inspected. The A261784 leading asymptotic was already posted; it is not claimed as a new prediction, and the entry is not said to label it a conjecture. Priority for the additional uniform results is not established by its bounded literature search. It separates the *proof* of a statement from *priority* for that statement, and says that no uninspected repository theorem is assumed and no Lean certification is inherited merely by association.

11.2 What is new, and a correction

Manuscript 01's abstract says that it proves the two conjectures posted in A261781, the recurrence order $k(k+1)/2$ of column k and the limit $d_{k+1} - d_k \rightarrow 1/L$; its delivered README and its OEIS draft (Section 22) say the same. That claim is narrowed here.

- A homogeneous constant-coefficient recurrence of order $\binom{k+1}{2}$ for column k , with the product denominator $Q_{k,1}(z) = \prod_{j=1}^k (2(1-z)^j - 1)$, is Proposition 25 of Munarini, Poneti and Rinaldi [4] (printed page 16), published in 2009, eight years before the conjecture; their Remark 26 prints the cases of two and three rows, the first of which is manuscript 01's example (2.8). Manuscript 01 cites Propositions 1, 12, 13 and 29 of that paper, but not Proposition 25 or Remark 26. What it proves that is new is the *minimality*: the product is the reduced denominator, so no eventual constant-coefficient recurrence of smaller order exists, even over $\mathbb{C}$. Manuscript 06 states this correctly (Section 1 and Remark 2.3).
- The limit $d_{k+1} - d_k \rightarrow 1/L$ is immediate from the closed form of [4, Proposition 13], as both manuscripts say. Manuscript 01's genuine strengthening is that the increments *increase strictly*

and stay below $1/L$ (Theorem 12.1).

- Manuscript 01 says that Louchard’s paper [6] was identified through A261780 but not retrieved. That remains a true statement about its own preparation and its non-claim stands; but the paper is retrievable, and manuscript 06 inspected the author’s long version and found no matching coverage-window statement (Section 1). The intake confirmed that the author’s publication list carries it as item 99 but did not read it.

The generating functions and the positive Stirling–Fubini transform are Munarini, Poneti and Rinaldi’s, and both manuscripts credit them; the leading A261784 law, d_* and the decimal value of c_* are Kotěšovec’s [3].

New in manuscript 01 relative to manuscript 06, and printed below: strict monotonicity of the increments (Section 12); every fixed order of the coverage probability (Section 13); the entire Stirling kernel with its differential recurrence, its third polynomial and a second exact algorithm (Section 14); a recursion for the cumulants (Section 15); the Poisson law of the Stirling defect (Section 16); the third diagonal correction and fifty-digit constants (Section 17); an asymptotic inversion with integer threshold brackets, which is an instance of the repository’s inversion apparatus (Section 18); exact-ratio coverage and defect tables (Sections 16 and 19); and its research questions and proof audit (Sections 20–22).

11.3 Manuscript 01’s second routes to Part I’s results

Part I	Manuscript 01’s version and route
(2.1)–(2.3), (2.4)	the same classical identities at $u = 1$, credited to [4]; the Stirling transform proved by the same geometric expansion and finite difference
Theorem 2.2	the same theorem at $u = 1$, same proof (Remark 2.3); the recurrence written from $n > D$
(2.9), Corollary 2.5	the same constants and expansions of d_k , c_k ; adds strict monotonicity (Theorem 12.1); lacks (2.11)
Theorem 4.1	the bound $(k - 1)\rho^{n+1}$, $u = 1$ only (Remark 4.2)
(5.4), Corollary 5.3	the same expansion of $\mathbb{P}(M = 0)$ and Poisson limit, by Bonferroni truncation of inclusion–exclusion (note after Theorem 5.1); no generating function in complex w , no probability-mass or total-variation statement
(7.5)	the same lemma, same proof
(7.7), Lemma 7.2	the untruncated kernel W_n , with no hypothesis on k (Remark 7.3, Section 14)
Theorem 7.1, (7.16), (7.17)	the same theorem and prescription; tails by $ W_n(w) \leq e^{ w }$ (notes after the proof of Theorem 7.1 and after (7.17))
Corollary 7.4, (9.1)	the same constants and β_1 , β_2 ; adds β_3 and all orders (Section 17)
ED, Var D in Corollary 7.5	implied by the Poisson law of $\Delta_{n,k} = D$ (Theorem 16.2)
Table 3, Table 4	Tables 8 and 7, from its own exact counts; equal where they overlap

Table 5: Results of Part I that manuscript 01 proves as well, and its version of each. None is reprinted in Part II.

Results of Part I that manuscript 01 does not have: the column weight u throughout; the initial impulse (Remark 2.4); the whole Hankel section (Theorem 3.1); the uniform constant of Theorem 4.1; the coverage generating function, the probability-mass and total-variation expansions

and the moving-mean remark of Section 5; the joint Gaussian–Poisson law of Section 6; the column central limit theorem on dense rays (Corollary 7.5); and the prescribed-column threshold of Section 8.

12 Strict monotonicity of the growth-constant increments

Extend d_k to real $x > 0$ by $d(x) = (1 - 2^{-1/x})^{-1}$, so that $d_k = d(k)$.

Theorem 12.1 (Strengthened increment statement). *The sequence $d_{k+1} - d_k$, $k \geq 1$, is strictly increasing, is always below $1/L$, and tends to $1/L$.*

Proof. Put $t = L/(2x)$. Since $d(x) = \frac{1}{2} + \frac{1}{2} \coth t$,

$$d'(x) = \frac{1}{L} \left(\frac{t}{\sinh t} \right)^2. \quad (12.1)$$

This lies strictly between 0 and $1/L$. The function $t/\sinh t$ is strictly decreasing for $t > 0$, because $\sinh t - t \cosh t < 0$; the latter expression vanishes at 0 and has derivative $-t \sinh t < 0$. As x increases, t decreases; thus d' increases strictly to $1/L$. Integrating d' over $[k, k+1]$ proves the claim. $\square$

Remark 12.2 (Further details from manuscript 01). (i) The inverse of (2.3) is $A(n, k) = \sum_{j=0}^k \binom{k}{j} T(n, j)$: choose the nonzero rows. Also $T(n, k) = 0$ for $k > n$, and the rational expression (2.1) at $u = 1$ equals 1 for $k = 0$, as it must. (ii) The bijection with the sorted-word compositions of A261780 and A261781 lists the row label i as many times as the entry in row i of a column, in increasing order; column boundaries become part boundaries, and the construction is reversible. (iii) At the unique smallest-modulus pole $1 - a_k$ only the $j = k$ summand of (2.3) contributes, with residue $-a_k/(2k)$, so its coefficient contribution is $\frac{a_k}{2k}(1 - a_k)^{-n-1} = c_k d_k^n$. (iv) The alternating sum (2.3) explains why cancellation is a numerical concern when all rows must be occupied; the positive transform (2.4) is an exact computational alternative to it.

13 Every fixed order in the coverage window

For real $x \rightarrow \infty$ the exact constants have the expansions

$$\log d(x) = \log \frac{x}{L} + \frac{L}{2x} - \frac{L^2}{24x^2} + O(x^{-4}), \quad (13.1)$$

$$\log c(x) = -\log(2L) - \frac{L}{2x} - \frac{L^2}{24x^2} + O(x^{-4}), \quad (13.2)$$

where $c(x) = (d(x) - 1)/(2x)$; these are the identities for $\log d_m$ and $\log c_m$ in the proof of Theorem 5.1, with $g_0(x) = -L^2 x^2/24 + O(x^4)$, and both differences from the leading logarithms are analytic in $1/x$ at 0. Let $\mathcal{T}_a(x)$ denote the Touchard polynomial, defined by $\sum_{j \geq 0} j^a x^j / j! = e^x \mathcal{T}_a(x)$.

Theorem 13.1 (All fixed orders for coverage). *Let $k \rightarrow \infty$ through integers, with $h = n/k = \log k + s$ and $\eta = ke^{-h} = e^{-s}$, where s stays in a fixed compact real interval. For every fixed $J \geq 0$ there are polynomials $\mathcal{Q}_1, \dots, \mathcal{Q}_J$ in h and η , with coefficients polynomial in L over $\mathbb{Q}$, such that*

$$\frac{T(n, k)}{A(n, k)} = \mathbb{P}(M = 0) = e^{-\eta} \left[1 + \sum_{j=1}^J \frac{\mathcal{Q}_j(h, \eta; L)}{k^j} + O\left(\frac{(1+h)^{J+1}}{k^{J+1}}\right) \right], \quad (13.3)$$

uniformly in such s . Each $\mathcal{Q}_j$ has degree at most j in h , and

$$\mathcal{Q}_1(h, \eta; L) = \frac{1}{2}((1-L)h\eta - (h+1)\eta^2).$$

For $J = 1$ this is (5.4).

Proof. Put $\epsilon = 1/k$ (a local letter; Part I's ϵ in Section 7.4 is $1/n$). Using the analytic expansions behind (13.1)–(13.2), form the formal power series in ϵ

$$\Psi(j, h, \epsilon) = \sum_{i=0}^{j-1} \log(1 - i\epsilon) + \log \frac{c(\epsilon^{-1} - j)}{c(\epsilon^{-1})} + \frac{h}{\epsilon} \left(\log \frac{d(\epsilon^{-1} - j)}{d(\epsilon^{-1})} + j\epsilon \right). \quad (13.4)$$

By (5.2) and Theorem 4.1, $\mathbb{E}(M)_j = \eta^j \exp \Psi(j, h, 1/k)$ up to the exponentially small spectral errors. The cancellation in the last parentheses removes its linear leading term. Every coefficient of Ψ is a polynomial in j , h , L , and its dependence on h is at most linear. Hence $[\epsilon^i]e^\Psi$ is a polynomial in j , h , L of degree at most i in h . Replace each monomial j^a in that coefficient by $\mathcal{T}_a(-\eta)$ to obtain $\mathcal{Q}_i$; this is the summation $\sum_j (-\eta)^j j^a / j! = e^{-\eta} \mathcal{T}_a(-\eta)$ of the inclusion–exclusion series $\mathbb{P}(M = 0) = \sum_j (-1)^j \mathbb{E}(M)_j / j!$.

To justify the operation, truncate at $j \leq C \log k$. Taylor's formula applied to the finitely many logarithms and to (13.1)–(13.2), continued to the required order, bounds the moment remainder by

$$\frac{\eta^j}{k^{J+1}} C_J (1+h)^{J+1} (1+j)^{C'_J}$$

for constants C_J , C'_J on the chosen s -interval. The exponential of the small discarded logarithm is uniformly bounded in this range. Summing against $1/j!$ preserves the stated error, and the Bonferroni inequalities bound the tail superalgebraically, as in the second route noted after Theorem 5.1. This proves both the algorithm and the uniform remainder. $\square$

The appearance of $\log k$ inside the coefficients, through h , is intrinsic to the transition scale. Theorem 13.1 is an all-orders Poincaré expansion of the single probability $\mathbb{P}(M = 0)$; it does not assert convergence as $J \rightarrow \infty$, nor a classification of exponentially small terms, and it does not extend Part I's expansion of the whole law of M beyond the first correction (see the note to Research question 2 in Section 10).

14 The entire Stirling kernel

For $n \geq 1$ define, with the falling factorial $(n)_\ell$ of Part I and $(n)_0 = 1$,

$$\chi_{n,\ell} = \frac{\left[\begin{smallmatrix} n \\ n-\ell \end{smallmatrix} \right]}{(n)_\ell^2}, \quad \mathbf{W}_n(w) = \sum_{\ell=0}^{n-1} \chi_{n,\ell} w^\ell, \quad (14.1)$$

and set $\chi_{n,\ell} = 0$ outside $0 \leq \ell < n$. By Remark 7.3, Part I's kernel is the truncation $H_{n,k}(z) = \sum_{\ell \leq n-k} \chi_{n,\ell} (Lz)^\ell$.

Proposition 14.1 (Positive coefficient representation). *Let $f(z) = e^z - 1$ and*

$$\tilde{T}(n, k) = \frac{n!}{2L^{n+1}} [z^n] f(z)^k \mathbf{W}_n(Lz). \quad (14.2)$$

For $n \geq k \geq 1$, $T(n, k) = \tilde{T}(n, k)(1 + O_\vartheta(\vartheta^k))$.

Proof. In (2.4) replace $\mathbf{F}_q$ using (7.5), use $\{q_k\} = q! [z^q] f(z)^k / k!$ and put $q = n - \ell$. The resulting exact main sum is

$$\frac{1}{2n! L^{n+1}} \sum_{\ell=0}^{n-1} \left[\begin{smallmatrix} n \\ n-\ell \end{smallmatrix} \right] ((n-\ell)!)^2 L^\ell [z^{n-\ell}] f(z)^k,$$

which is (14.2). Positivity of every summand transports the relative error. $\square$

Lemma 14.2 (Bounds and an exact differential recurrence). *For $0 \leq \ell < n$, $0 \leq \chi_{n,\ell} \leq 1/\ell!$; consequently $|\mathbf{W}_n(w)| \leq e^{|w|}$ for every complex w . With $D_w = w \, d/dw$,*

$$(n+1)^2 \mathbf{W}_{n+1}(w) = ((n+1 - D_w)^2 + nw) \mathbf{W}_n(w), \quad \mathbf{W}_1(w) = 1. \quad (14.3)$$

Proof. The first-kind coefficient $\begin{bmatrix} n \\ n-\ell \end{bmatrix}$ is the ℓ th elementary symmetric polynomial of $1, 2, \dots, n-1$. Each of its $\binom{n-1}{\ell}$ summands is at most $(n-1)_\ell$. Hence

$$\chi_{n,\ell} \leq \frac{(n-1)_\ell^2}{\ell! (n)_\ell^2} \leq \frac{1}{\ell!}.$$

The first-kind recurrence gives

$$\chi_{n+1,\ell} = \frac{(n+1-\ell)^2}{(n+1)^2} \chi_{n,\ell} + \frac{n}{(n+1)^2} \chi_{n,\ell-1}.$$

Multiplying by w^ℓ and summing proves (14.3). $\square$

Theorem 14.3 (Uniform entire kernel expansion). *For each fixed $J \geq 0$ there are rational polynomials $P_0, \dots, P_J$, with $P_0 = 1$ and $\deg P_r \leq 2r$, such that*

$$\mathbf{W}_n(w) = e^{w/2} \left(\sum_{r=0}^J \frac{P_r(w)}{n^r} + O_{J,K}(n^{-J-1}) \right) \quad (14.4)$$

uniformly on every compact $K \subset \mathbb{C}$, with no condition relating n to any k . The expansion may be differentiated any fixed number of times, uniformly on compact sets. The polynomials are those of (7.15):

$$\begin{aligned} P_1(w) &= \frac{w^2}{12} - \frac{w}{2}, & P_2(w) &= \frac{w^4}{288} - \frac{w^2}{12}, \\ P_3(w) &= \frac{w^6}{10368} + \frac{w^5}{576} + \frac{w^4}{480} - \frac{w^3}{24} - \frac{w^2}{12}. \end{aligned} \quad (14.5)$$

Proof. This is a coefficient-level proof with a uniform remainder. Let $\Sigma_j(n) = \sum_{a=1}^{n-1} a^j$, regarded as its Faulhaber polynomial in n . With $\epsilon = 1/n$ put $\mathbf{a}_j(\epsilon) = \epsilon^{2j} \Sigma_j(1/\epsilon)$. This is a polynomial beginning in degree $j-1$, and $\mathbf{a}_1(\epsilon) = (1-\epsilon)/2$. For a fixed integer $\ell \geq 0$ define

$$\mathbf{e}_\ell(\epsilon) = [t^\ell] \exp \left(\sum_{j=1}^{\ell} \frac{(-1)^{j-1}}{j} \mathbf{a}_j(\epsilon) t^j \right), \quad \mathbf{r}_\ell(\epsilon) = \mathbf{e}_\ell(\epsilon) \prod_{a=0}^{\ell-1} (1 - a\epsilon)^{-2}. \quad (14.6)$$

Newton's identity for elementary symmetric functions shows that $\mathbf{r}_\ell(1/n) = \chi_{n,\ell}$ whenever $n > \ell$. In particular $\mathbf{r}_\ell(0) = 1/(2^\ell \ell!)$.

The exponent in (14.6) is $t/2 + \sum_{r \geq 1} \epsilon^r \mathbf{b}_r(t)$ with $\deg \mathbf{b}_r \leq r+1$. Thus at total order ϵ^r the part multiplying $e^{t/2}$ has degree at most $2r$ in t , and extracting t^ℓ converts $t^q e^{t/2}$ into $2^q (\ell)_q / (2^\ell \ell!)$. The other factor of $\mathbf{r}_\ell$ has logarithm $2 \sum_{r \geq 1} \frac{\epsilon^r}{r} \sum_{a=0}^{\ell-1} a^r$, whose coefficient at order r is a polynomial in ℓ of degree $r+1 \leq 2r$. Therefore

$$[\epsilon^r] \mathbf{r}_\ell(\epsilon) = \frac{\Pi_r(\ell)}{2^\ell \ell!}, \quad \Pi_r \in \mathbb{Q}[\ell], \quad \deg \Pi_r \leq 2r; \quad (14.7)$$

for example $\Pi_0 = 1$ and $\Pi_1(\ell) = \ell(\ell-4)/3$, the polynomial $d(d-4)/3$ of the proof of Lemma 7.2.

Next, the Taylor remainders are bounded uniformly in ℓ . There are absolute $c, C > 0$ such that, for $|\epsilon| \leq c/(\ell+1)^2$ and $2 \leq j \leq \ell$,

$$|a_j(\epsilon)| \leq C|\epsilon|^{j-1}. \quad (14.8)$$

For completeness: Faulhaber's formula writes a_j as $\frac{1}{j+1} \sum_{a=0}^j \binom{j+1}{a} \mathbf{B}_a \epsilon^{j-1+a}$, with Bernoulli numbers $\mathbf{B}_a$. The coefficients of $t/(e^t - 1)$ obey $|\mathbf{B}_a|/a! \leq C_0$ by a Cauchy bound on a fixed circle of radius 1. Since $j!/(j+1-a)! \leq j^{a-1}$ for $a \geq 1$, the sum of the terms after the first is bounded by a geometric series in $j|\epsilon|$.

For $\ell \geq 1$, apply Cauchy's coefficient bound in (14.6) on $|t| = 2\ell$. The $j = 1$ term contributes at most $\ell + O(1)$ to the real part of the exponent, while (14.8) bounds the sum of the terms $j \geq 2$ by a constant, because $|\epsilon|\ell^2$ is bounded; the product $\prod(1 - a\epsilon)^{-2}$ is bounded for the same reason. Stirling's factorial estimate then gives

$$\sup_{|\epsilon| \leq c/(\ell+1)^2} |r_\ell(\epsilon)| \leq \frac{C\sqrt{\ell+1}}{2^\ell \ell!}; \quad (14.9)$$

for $\ell = 0$ this is immediate. Cauchy's Taylor-remainder estimate at $\epsilon = 1/n$ consequently yields

$$\chi_{n,\ell} = \frac{1}{2^\ell \ell!} \sum_{r=0}^J \frac{\Pi_r(\ell)}{n^r} + O_J\left(\frac{(\ell+1)^{2J+5/2}}{2^\ell \ell! n^{J+1}}\right) \quad (14.10)$$

whenever $\ell \leq \delta_0 \sqrt{n}$, for a sufficiently small absolute $\delta_0 > 0$ and all sufficiently large n .

Multiply (14.10) by w^ℓ and sum. On $|w| \leq V$ the remainder is summable uniformly. Beyond $\delta_0 \sqrt{n}$ the bound $\chi_{n,\ell} \leq 1/\ell!$ makes the original tail smaller than every power of n^{-1} , and the same holds for the polynomially weighted Taylor coefficients in (14.7). Hence

$$\mathbf{W}_n(w) = \sum_{r=0}^J \frac{1}{n^r} \sum_{\ell \geq 0} \frac{\Pi_r(\ell)}{\ell!} \left(\frac{w}{2}\right)^\ell + O_{J,V}(n^{-J-1}).$$

The inner sum is $e^{w/2}$ times a polynomial of degree at most $2r$, by the Touchard identity. This proves (14.4); expanding (14.6) gives P_1, P_2, P_3 . Cauchy's derivative estimate on a slightly larger compact disk gives the differentiated expansions. $\square$

A finite exact coefficient algorithm. The polynomials can be computed without a symbolic parameter ℓ . For a requested order J , compute $e_\ell(n) = \lfloor \frac{n}{n-\ell} \rfloor$ for $0 \leq \ell \leq 2J$ by

$$e_0(n) = 1, \quad \ell e_\ell(n) = \sum_{j=1}^{\ell} (-1)^{j-1} e_{\ell-j}(n) \Sigma_j(n), \quad (14.11)$$

expand $e_\ell(n)/(n)_\ell^2$ in powers of $1/n$ to order J , and use $\deg P_r \leq 2r$:

$$P_r(w) = \sum_{m=0}^{2r} w^m \sum_{\ell=0}^m \frac{(-1/2)^{m-\ell}}{(m-\ell)!} [n^{-r}] \frac{e_\ell(n)}{(n)_\ell^2}. \quad (14.12)$$

Here $[n^{-r}]$ is the coefficient in the expansion at infinity, not a coefficient of the polynomial numerator alone. Manuscript 01's program (`code/01-two-poisson-verify.py`) runs (14.11)–(14.12) exactly over $\mathbb{Q}$ and checks (14.3) coefficientwise through $n = 80$. This is a second finite route to Part I's algorithm (7.12)–(7.14). [Added 3 October 2026, batch 85B: the intake checked P_3 numerically: at $w = 1.3$, $n^3(e^{-w/2}\mathbf{W}_n(w) - 1 - P_1/n - P_2/n^2)$ is $-0.22070, -0.22010, -0.21981$ for $n = 400, 800, 1600$, against $P_3(1.3) = -0.21951$, with the expected $O(1/n)$ approach.]

15 Complements on compact rays

Manuscript 01 writes the leading factor of Theorem 7.1 as

$$\mathcal{B}(n, k) = \frac{n! f(r)^k e^v}{2L^{n+1} r^n \sqrt{2\pi k b}}, \quad T(n, k) = \mathcal{B}(n, k) \left[1 + \sum_{j=1}^J \frac{E_j(\lambda)}{k^j} + O_{J,K}(k^{-J-1}) \right]. \quad (15.1)$$

It adds two observations and a recursion.

The variance b . $b = \kappa_2 = \lambda(1 + r - \lambda)$ is the variance of the positive-integer random variable with probabilities proportional to $r^j/j!$, $j \geq 1$, which explains its positivity; the proof needs only the monotonicity of $r/(1 - e^{-r})$.

A recursion for the cumulants. With independent variables z, a , define

$$\mathcal{K}_1(z, a) = a, \quad \mathcal{K}_{j+1} = z \partial_z \mathcal{K}_j + a(1 + z - a) \partial_a \mathcal{K}_j. \quad (15.2)$$

Then $\kappa_j = \mathcal{K}_j(r, \lambda)$. This follows from $\mathcal{D}\lambda = \lambda(1 + z - \lambda)$ for $\lambda(z) = \mathcal{D} \log f(z) = z/(1 - e^{-z})$, before evaluation at the saddle. One must not differentiate a specialized formula while holding λ fixed. (For example $\mathcal{K}_3 = a(2a^2 - 3az - 3a + z^2 + 3z + 1)$, which is the expanded κ_3 printed after Theorem 7.1.) Together with (14.12) and (7.16), this specifies every E_j by finite rational operations and the evaluation of r ; in fact E_j is a rational function of r, λ, L whose denominators are powers of b and λ .

Remark 15.1 (Floors and endpoints). The ratio in Theorem 7.1 is the *actual* n/k . For $n = \lfloor \lambda_0 k \rfloor$, replacing it prematurely by λ_0 inside exponential factors can lose a nonvanishing factor. The theorem is not uniform as $\lambda \downarrow 1$ or $\lambda \rightarrow \infty$; at the boundary $n = k$ the exact identity is $T(n, n) = \mathbf{F}_n$, and the coverage window, where $\lambda \sim \log k$, needs the separate proof of Section 5. Manuscript 01's optional program `code/01-two-poisson-ray_spotchecks.py` tests the expansion at $\lambda = 3/2, 3$ and 5 for $40 \leq k \leq 200$ with 75-digit arithmetic (`data/01-two-poisson-ray_spot_checks.csv`); the relative error after the third correction is between 2.38×10^{-12} and 1.63×10^{-9} in absolute value at the seven tested points.

16 The Stirling defect: a second Poisson limit

The positive identity (2.4) supplies an auxiliary probability space. It should not be confused with the zero-row statistic M , which vanishes identically on the objects counted by T .

Definition 16.1. For $n \geq k \geq 1$, define $Q_{n,k}$ on $\{k, \dots, n\}$ by

$$\mathbb{P}(Q_{n,k} = q) = \frac{\begin{bmatrix} n \\ q \end{bmatrix} \{q\} \mathbf{F}_q}{\sum_{p=k}^n \begin{bmatrix} n \\ p \end{bmatrix} \{p\} \mathbf{F}_p}, \quad \Delta_{n,k} = n - Q_{n,k}. \quad (16.1)$$

$\Delta_{n,k}$ is the *Stirling defect* of the transform. $Q_{n,k}$ and $\Delta_{n,k}$ are Part I's Q and D of Section 7.6.

Theorem 16.2 (Poisson defect and analytic generating functions). *Suppose $n, k \rightarrow \infty$ and $n/k \rightarrow \lambda_0 \in (1, \infty)$. Then*

$$\Delta_{n,k} \xrightarrow{d} \text{Poisson}\left(\frac{Lr(\lambda_0)}{2}\right). \quad (16.2)$$

More precisely, uniformly for complex ξ in compact sets and for n/k in compact subsets of $(1, \infty)$,

$$\mathbb{E}\xi^{\Delta_{n,k}} = e^{v(\xi-1)}(1 + O(k^{-1})), \quad v = \frac{Lr(n/k)}{2}. \quad (16.3)$$

There is an expansion through every fixed order, locally uniformly in ξ , and the same holds for each fixed derivative with respect to ξ .

Proof. Marking $n - q = \ell$ in the main transform replaces $W_n(Lz)$ by $W_n(\xi Lz)$. Lemma 14.2 and positivity bound the Fubini replacement error, also for bounded complex ξ by using absolute values. The proof of Theorem 7.1, with ξ in a compact set, now replaces $e^{Lr/2}$ by $e^{\xi Lr/2}$; this uses Theorem 14.3 on compact sets of $\mathbb{C}$. Division by the unmarked coefficient gives (16.3); the denominator's leading factor is positive and bounded away from zero after normalization, so the division is uniform. Replacing $P_p(Lre^{\tau y})$ by $P_p(\xi Lre^{\tau y})$ in (7.16) gives every higher order. Uniform holomorphy on a larger compact set permits differentiation. Coefficient extraction at $\xi = 0$ yields convergence of each probability to the corresponding Poisson mass, and these limiting masses sum to one. $\square$

Differentiating (16.3) once and twice at $\xi = 1$ gives $\mathbb{E}\Delta_{n,k} = v + O(k^{-1})$ and $\text{Var } \Delta_{n,k} = v + O(k^{-1})$, the two moments of Corollary 7.5. The theorem explains why only a bounded distance below the top Stirling index contributes at leading order, although the positive sum ranges from k to n : the factor $e^v = e^{Lr/2}$ in (15.1) is exactly the aggregate contribution of the limiting Poisson cloud. The report does not assert that $\Delta_{n,k}$ is a canonical statistic on unadorned matrices; finding such a statistic is a separate combinatorial problem (Section 20).

For $n = 2N$, $k = N$ the limiting mean is $v = 0.55230808135934145323\dots$. Direct evaluation of the positive weights in (16.1) gives Table 6.

ℓ	$N = 10$	$N = 50$	$N = 200$	$e^{-v}v^\ell/\ell!$
0	0.592485680	0.578956893	0.576452261	0.575619699
1	0.314681461	0.317313214	0.317769736	0.317919411
2	0.078997588	0.086058333	0.087361728	0.087794730
3	0.012377926	0.015394446	0.015970554	0.016163246
4	0.001344681	0.002042736	0.002184003	0.002231773
5	0.000106192	0.000214396	0.000238309	0.000246525

Table 6: Convergence of $\mathbb{P}(\Delta_{2N,N} = \ell)$ to its Poisson limit (manuscript 01).

17 A261784: the third correction and every further one

Corollary 17.1 (Diagonal expansion to every order). *For every fixed $J \geq 0$,*

$$a_N = c_* d_*^N \frac{(N!)^2}{N} \left(1 + \sum_{j=1}^J \frac{\beta_j}{N^j} + O_J(N^{-J-1}) \right). \quad (17.1)$$

Every β_j is a rational function of L and r , specified exactly by the prescription (7.16) (with (15.2) and (14.12)) and the central binomial expansion. Besides $\beta_1 = \beta$ and β_2 of Part I,

$$\beta_3 = -0.00244353842284859500374038095590013033570627782694\dots \quad (17.2)$$

Proof. At $\lambda = 2$ one has $f(r) = r/(2 - r)$ and $b = 2(r - 1)$. Substitute these in (15.1) and use

$$\frac{(2N)!}{(N!)^2} = \frac{4^N}{\sqrt{\pi N}} \left(1 - \frac{1}{8N} + \frac{1}{128N^2} + \frac{5}{1024N^3} + O(N^{-4}) \right). \quad (17.3)$$

With $E_j = E_j(2)$ this gives

$$\beta_1 = E_1 - \frac{1}{8}, \quad \beta_2 = E_2 - \frac{E_1}{8} + \frac{1}{128}, \quad \beta_3 = E_3 - \frac{E_2}{8} + \frac{E_1}{128} + \frac{5}{1024}, \quad (17.4)$$

the first two being (9.1). The same multiplication determines every further coefficient. All computations are finite; the displayed decimals are evaluations of these exact prescriptions, not fitted constants. $\square$

Manuscript 01's ninety-digit evaluation gives, truncated to fifty decimals,

$$\begin{aligned} r &= 1.59362426004004009232304187587516024178900242481885\dots, \\ d_* &= 12.85568917366612131932660300054233483234414963116428\dots, \\ c_* &= 0.25886492311146555025177523170232718427705044811049\dots, \\ E_1(2) &= -0.39680670155648266388914410306946945484974842970590\dots, \\ E_2(2) &= 0.07087549039414544547365369135067624979402488769624\dots, \\ E_3(2) &= 0.00463313773232960649210026876816463100456049274216\dots, \\ \beta_1 &= -0.52180670155648266388914410306946945484974842970590\dots, \\ \beta_2 &= 0.12828882808870577845979670423435993165024344140947\dots \end{aligned}$$

[Added 3 October 2026, batch 85B: the intake compared these with manuscript 06's independent seventy-digit values in `data/06-hankel-diagonal_constants.json`; $r, b, v, d_*, c_*, \beta_1$ and β_2 agree in every printed digit.]

Let $a_N^{[i]}$ denote (17.1) truncated after β_i/N^i , with $i = 0$ the leading approximation. Table 7 reports $a_N^{[i]}/a_N - 1$, using exact integer a_N and 90-digit working precision, and Figure 2 plots the same quantities for more N . (Part I's Table 4 uses the reciprocal convention.) The very small

N	$i = 0$	$i = 1$	$i = 2$	$i = 3$
10	$+5.363 \times 10^{-2}$	-1.350×10^{-3}	$+2.063 \times 10^{-6}$	-5.117×10^{-7}
20	$+2.645 \times 10^{-2}$	-3.289×10^{-4}	$+2.844 \times 10^{-7}$	-2.917×10^{-8}
50	$+1.049 \times 10^{-2}$	-5.183×10^{-5}	$+1.905 \times 10^{-8}$	-7.075×10^{-10}
100	$+5.232 \times 10^{-3}$	-1.289×10^{-5}	$+2.413 \times 10^{-9}$	-4.344×10^{-11}
200	$+2.613 \times 10^{-3}$	-3.215×10^{-6}	$+3.035 \times 10^{-10}$	-2.691×10^{-12}

Table 7: Relative errors of successive asymptotic approximations to A261784 (manuscript 01).

third-order residual at $N = 200$ is consistent with the proved $O(N^{-4})$ relative remainder; between $N = 100$ and $N = 200$ it falls by a factor of about 16. It does not by itself establish that order, which comes from the uniform analytic proof.

18 Asymptotic inversion and integer thresholds

[Added 3 October 2026, batch 85B: this section is an instance of the inversion apparatus of the repository's transseries volume [11]: Proposition 18.1 is a first step of its perturbed inversion,

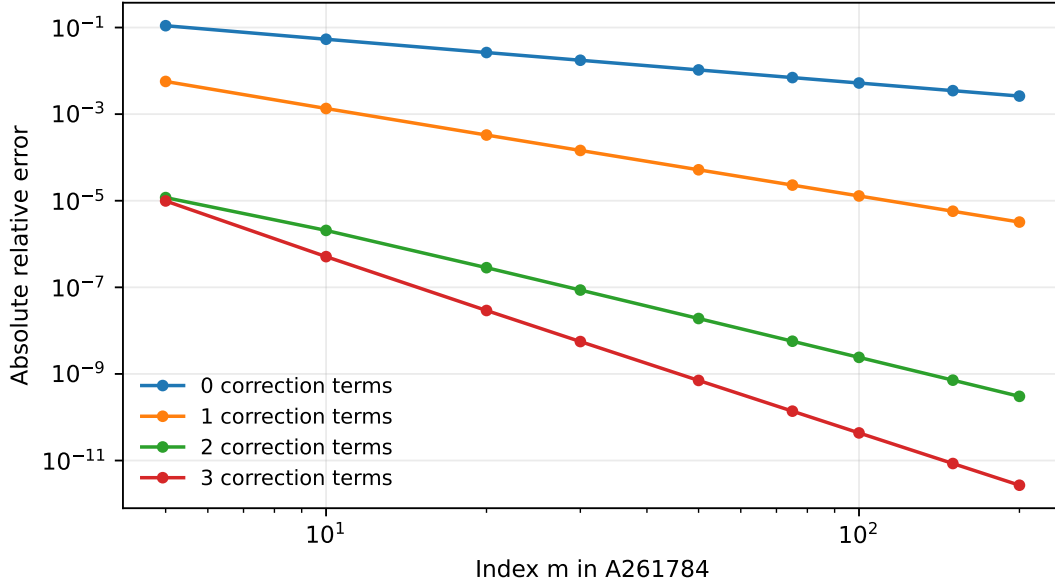


Figure 2: Absolute relative errors from the same exact data as Table 7, with additional intermediate indices (manuscript 01). These are numerical validations, not certified interval error bounds.

Theorem `p0:thm:perturbed-inversion`, for the core $2x \log(\alpha x)$, and Proposition 18.2 is a case of the separation condition of its staircase theorem, `p0:thm:staircase`. Manuscript 01 does not cite either; it was motivated by the volume’s overview README, read at its pin. No novelty is claimed for the method; the constants are specific to A261784.]

A rapidly growing sequence has a natural discrete inverse, but no unique smooth interpolation; the two objects are kept separate.

A selected smooth comparator. For fixed $J \geq 1$ and sufficiently large real x define

$$\mathcal{A}^{[J]}(x) = c_* d_*^x \frac{\Gamma(x+1)^2}{x} \left(1 + \sum_{j=1}^J \beta_j x^{-j} \right). \quad (18.1)$$

It is positive and eventually strictly increasing, because the derivative of its logarithm is $2 \log x + \log d_* + O(x^{-1}) > 0$. Set

$$\alpha = \frac{\sqrt{d_*}}{e}, \quad \tilde{\beta}_1 = \beta_1 + \frac{1}{6} = -0.35514003488981599722247743640280278818 \dots \quad (18.2)$$

($\tilde{\beta}_1$ is manuscript 01’s “ β_1 ”; it is not Part I’s β_1 .) Stirling’s expansion and (17.1) give, at integer N ,

$$\log a_N = 2N \log(\alpha N) + \log(2\pi c_*) + \frac{\tilde{\beta}_1}{N} + O(N^{-2}). \quad (18.3)$$

The cancellation of the $\log N$ term is specific to the $(N!)^2/N$ normalization.

Proposition 18.1 (Lambert- W inverse and first residual correction). *Let $Y \rightarrow \infty$ and put*

$$\Lambda = \frac{\log Y - \log(2\pi c_*)}{2}, \quad \varpi = W_0(\alpha \Lambda), \quad x_0 = \frac{\Lambda}{\varpi}. \quad (18.4)$$

If $x_J(Y)$ solves $\mathcal{A}^{[J]}(x_J) = Y$, then

$$x_J(Y) = x_0 - \frac{\tilde{\beta}_1}{2x_0(1+\varpi)} + O\left(\frac{x_0^{-2}}{\log x_0}\right). \quad (18.5)$$

Further terms are obtained by substituting a formal correction into the logarithm of (18.1) and cancelling successive residuals.

Proof. The definition of ϖ gives $\alpha x_0 = e^\varpi$, so $x_0 \log(\alpha x_0) = \Lambda$ and x_0 inverts the dominant core exactly. Substitution in the logarithm of (18.1) leaves the residual $\tilde{\beta}_1/x_0 + O(x_0^{-2})$. The derivative of the core is $2(1 + \log(\alpha x_0)) = 2(1 + \varpi)$, and the displayed correction cancels the residual to $O(x_0^{-2})$; quadratic errors in the correction are smaller. The derivative stays comparable to $\log x_0$, and the mean-value theorem transports the remaining logarithmic residual to the claimed root error. $\square$

The discrete threshold. Define

$$\mathcal{N}(Y) = \min\{N \geq 0 : a_N \geq Y\}. \quad (18.6)$$

The sequence a_N is strictly increasing. Indeed, pad every matrix counted by a_N with a new zero row and append two columns, each having a single 1 in that new row. This is an injection into the matrices counted by a_{N+1} . It is not surjective: a positive one-column matrix with $N+1$ rows and total $2N+2$ is not in its image.

Proposition 18.2 (Residual brackets for integer inversion). *For each fixed $J \geq 1$ there is a constant C_J such that, for all sufficiently large Y , if $\epsilon_J = C_J x_J(Y)^{-J-1} / \log x_J(Y)$, then*

$$\lceil x_J(Y) - \epsilon_J \rceil \leq \mathcal{N}(Y) \leq \lceil x_J(Y) + \epsilon_J \rceil. \quad (18.7)$$

In particular $\mathcal{N}(Y) = \Lambda/W_0(\alpha\Lambda) + O(1)$.

Proof. At integer N the logarithmic difference between a_N and $\mathcal{A}^{[J]}(N)$ is $O(N^{-J-1})$ by Corollary 17.1. The derivative of $\log \mathcal{A}^{[J]}$ is bounded below by a positive constant times $\log x_J$ in a fixed-width interval around x_J . Choosing C_J large enough, the model's change across ϵ_J dominates that logarithmic error at every integer in the interval. Integers strictly below $x_J - \epsilon_J$ cannot attain Y ; every integer at least $x_J + \epsilon_J$ does. Monotonicity handles integers farther from the interval. Taking ceilings gives (18.7). $\square$

No unconditional formula $\mathcal{N}(Y) = \lceil x_J(Y) \rceil$ follows. When the smooth root lies extremely close to an integer, even a tiny asymptotic error can decide which of two consecutive integer thresholds is correct; the bracket is a valid statement that does not hide that ambiguity. Its constants are existential consequences of the asymptotic proof; no numerically certified value of C_J is claimed.

19 Manuscript 01's checks

Manuscript 01's program `code/01-two-poisson-verify.py` (delivered as `scripts/verify.py`) uses Python integers for exact combinatorics, SymPy for rational polynomial operations, and mpmath at 90 decimal digits for evaluation. It (1) matches eight displayed rows of A261781 and all fifteen displayed terms of A261784, finite transcriptions marked as such in the source; (2) compares the positive Stirling transform with a separately implemented generating-function recurrence followed by inclusion–exclusion for every $0 \leq k \leq n \leq 40$; (3) computes the reduced

rational functions for $1 \leq k \leq 8$, checks the denominator degrees and coprimality, and verifies later terms against the extracted recurrence; (4) checks every coefficient of (14.3) through $n = 80$, constructs P_1, P_2, P_3 over $\mathbb{Q}$, and compares the finite Gaussian algorithm's E_1 with the closed formula (7.4); and (5) computes $a_0, \dots, a_{200}$ exactly (`data/01-two-poisson-a261784_exact.txt`) and produces the diagonal, coverage and defect tables, which are floating-point tables, not interval-arithmetic certificates. Its record (`data/01-two-poisson-verification.json`) says “ALL CHECKS PASSED”. Finite tests do not prove the infinite statements; they check algebra, indexing, normalization and implementation.

The coverage table takes n to be the nearest integer to $k \log k$ and uses the actual $h = n/k$ and $\eta = ke^{-h}$ in both approximations; the probabilities in its third column are ratios of exact integers. The correction is not advertised as monotone in accuracy for every small input: the theorem is a

k	n	Exact coverage	$e^{-\eta}$	First correction
10	23	0.314344674	0.366928440	0.319053131
20	60	0.339714115	0.369449445	0.341284787
40	148	0.355320143	0.371970377	0.355818395
80	351	0.360509443	0.369892970	0.360669046
120	574	0.359614880	0.366349652	0.359697754

Table 8: The first correction of (5.4) near the centre of the coverage transition (manuscript 01). Rounding n explains why the leading values are not all exactly e^{-1} .

uniform asymptotic statement for bounded s , not an approximation inequality for all k .

[Added 3 October 2026, batch 85B: at the intake's rerun on a copy, `verify.py -max-m 200` passed all checks in 162s (its README says about 16s), with every output byte-identical to the shipped one or equal apart from line endings and elapsed time; `ray_spotchecks.py`, which imports `verify` and so needs SymPy as well, reproduced its CSV byte for byte in 207s; and the figure program reproduced the PNG pixel for pixel. The 201 computed terms of A261784 were not compared with the OEIS b-file [8], and manuscript 01 does not claim that they were.]

20 Further research questions of manuscript 01

Manuscript 01 orders these by proximity to its methods, not by a claim that any is easy or guaranteed to be new. The bracketed notes re-scope them against Part I.

Uniformity between compact rays and the coverage window. Prove a single relative asymptotic valid on an expanding interval such as $1 + \varepsilon \leq n/k \leq \log k + C$, with an error that remains informative when coverage is exponentially rare. The kernel suggests extending (14.4) from fixed w to growing $w = Lr$, while the saddle must be controlled as $r \rightarrow \infty$; a successful result should reproduce the polynomial corrections of Theorem 13.1, not only the limiting curve. A concrete first step is a bound for the kernel remainder on $|w| \leq C \log n$ that keeps its exponential dependence on $|w|$. [Added: this is the second deliverable of Research question 1 of Part I; open.]

The near-diagonal boundary. At $n = k$ the answer is F_n , whereas the saddle variance b vanishes as $n/k \downarrow 1$. Determine expansions uniform when $\ell = n - k$ is fixed, when $\ell \rightarrow \infty$ with $\ell = o(k)$, and across their transition. The positive Stirling sum has only $\ell + 1$ summands in the fixed- ℓ regime, so exact polynomial descriptions may be feasible before any saddle approximation; matching them to a

large- ℓ saddle expansion would complete the present analysis at the endpoint. [Added: this is the first deliverable of Research question 1 of Part I; open.]

A combinatorial realization of the latent defect. Find a direct decorated-matrix construction whose forgetful map induces exactly the weights (16.1). Ideally the defect would count a familiar local feature and make the factor $e^{Lr/2}$ visible; such a construction could also explain the polynomials P_r and the negative first-order term in P_1 without Faulhaber expansion. [Added: open; both manuscripts decline to claim a canonical statistic.]

Joint laws with the number of columns. Mark the columns by u ; before requiring nonzero rows the bivariate generating function is $F_k(z; u)$ of (2.1). Study the joint distribution of the column count and the zero-row count in the coverage window, and of the column count together with a combinatorial defect on compact rays; the principal pole is governed by $\log(1 + 1/u)$, and uniform control near $u = 1$ is needed before claiming a central limit theorem or asymptotic independence. [Added 3 October 2026, batch 85B: the first half is answered by Part I. Theorem 4.1 gives the uniform control in u on compact intervals, Lemma 6.1 the Poisson law under column weights, and Theorem 6.2 the joint limit of the normalized column count and the zero-row count, with independence and the conditional Gaussian law given $M = m$. On compact rays Corollary 7.5 gives the column central limit theorem and its proof the exact mixture over $Q_{n,k}$; the joint law of the column count and the defect $\Delta_{n,k}$, or a combinatorial defect, remains open, as do the local refinements of Research question 3.]

Weighted rows, columns, and restricted alphabets. Replace the identical row weights or the unrestricted column entries by a specified weight system, and determine which features of the recurrence-order proof survive when distinct factors can share poles. The proof suggests a precise obstruction: cancellations and coincident spectra, rather than the mere presence of a rational generating function. A classification for finite sets of permitted column weights would be a useful structural extension. [Added: related to Research question 8 of Part I, which asks the same about the coverage law; open.]

Beyond all algebraic orders. The Fubini generating function has further poles at $L + 2\pi ij$. The present proofs bound their total effect exponentially but do not extract their separate saddle contributions. Derive those sectors, with their oscillatory phases, and compare their scale with the late terms of the algebraic expansion; this is where a genuinely multi-sector transseries, rather than a Poincaré series, should enter, and optimal truncation and rigorous remainder bounds would be part of that problem. [Added 3 October 2026, batch 85B: for the Fubini numbers themselves the sectors are exact and convergent, Theorem `q2:thm:fubini` of [11]; the open problem is to carry them through the Stirling transform and the saddle. The report on A260700 [10] lists the same open problem for its sequence.]

Effective certified computation and formalization. Make the constants in the compact-ray error bound and in (18.7) explicit; interval estimates on the central arc and a certified minor-arc gap could turn the inversion into a proof-producing threshold algorithm. The exact recurrence theorem is a suitable first formalization target: its core is pairwise disjoint simple poles and the equivalence between rational generating functions and eventual linear recurrences. The analytic kernel and saddle theorems are substantially larger tasks and should be scoped separately. [Added: the same target as Part I’s “practical order of attack”; nothing is formalized yet.]

21 Hypotheses that cannot be dropped

Manuscript 01 ends with a proof audit, kept here in full.

Constant versus polynomial coefficients. The minimality assertion is for homogeneous recurrences with constant coefficients. It is not a lower bound for recurrences with polynomial coefficients in n , nonlinear recurrences, or recurrences involving several columns k at once.

Nonnegative versus strictly positive entries. Entries are nonnegative; the restrictions are on entire rows and columns, not on each cell. Replacing cells by positive integers changes the generating functions and the sequence.

The polynomial part of a rational function. The fixed-spectrum formula for $A(n, k)$ applies to $n \geq 1$, and the recurrence for $T(n, k)$ starts after the numerator has ceased contributing. Ignoring either point creates incorrect initial conditions.

Positivity before relative error transport. The approximation of F_q is transported through the positive identity (2.4), not through the alternating inclusion–exclusion sum: an exponentially accurate approximation term by term in an alternating sum need not be relatively accurate after cancellation. The coverage calculation instead controls factorial moments and then uses Bonferroni truncation.

Compactness in the ray theorem. The uniform constants require n/k to stay away from 1 and infinity. Neither endpoint is covered by substituting it in (15.1), and the actual rational n/k must be used when n is rounded.

Two different Poisson variables. M counts zero rows in uniformly sampled matrices that may have zero rows; $\Delta_{n,k}$ is an auxiliary variable defined by positive transform weights. Their means, parameter regimes and sample spaces are different.

Formal Gaussian algebra. The prescription (7.16) means coefficient extraction first, Gaussian evaluation second; it does not require the integral of the untruncated exponential expression to converge.

Asymptotic expansion versus convergent series. Every order J is fixed before the size parameter tends to infinity. Neither growth estimates for the coefficients as $J \rightarrow \infty$ nor optimal truncation are asserted.

Numerics and proof status. Exact integer checks certify their finite inputs. The 90-digit numerical calculations are independent consistency checks, not rigorous floating-point enclosures or a machine-checked proof of the theorems.

22 Manuscript 01’s draft OEIS additions (not submitted)

Manuscript 01 includes draft additions to two OEIS entries, shipped as `data/01-two-poisson-OEIS_updates_draft.txt`. *They were not submitted*, by manuscript 01 or by this report;

manuscript 01 itself asks for independent review and a stable public manuscript link first, and for the earlier source and conjecture attributions to be preserved. Their substance, in this report's letters:

A261781. For $k > 0$ the reduced denominator of the ordinary generating function of column k is $\prod_{j=1}^k (2(1-x)^j - 1)$, so its minimal eventual homogeneous constant-coefficient recurrence has order $k(k+1)/2$: inclusion–exclusion expresses the column generating function as the alternating binomial sum of $(1-x)^j/(2(1-x)^j - 1)$, whose j th summand has simple poles at $1 - 2^{-1/j}e^{2\pi i\ell/j}$, and different summands have disjoint pole sets. The fixed-column constants are $c_k = 1/(2k(2^{1/k} - 1))$ and $d_k = 1/(1 - 2^{-1/k})$, and $d_{k+1} - d_k$ increases strictly to $1/\log 2$, with $d'(x)$ as in (12.1). Attribution should retain the existing conjecture attribution and cite the generating-function source.

[Added 3 October 2026, batch 85B: *before any use* this draft must be corrected as in Section 11.2. Its sentence saying that these statements prove the two claims labelled conjectures overstates them: a recurrence of order $k(k+1)/2$ with this denominator is [4, Proposition 25], with the case $k = 2$ in Remark 26, and the limit follows from Proposition 13 there; what is new is the minimality and the strict monotonicity. The shipped file is kept byte-identical and still contains the overstated sentence.]

A261784. With $L = \log 2$ and $r = 2 + W_0(-2e^{-2})$, the posted prefactor is $c_* = 2^{r/2}/(4\pi L\sqrt{r-1})$ and the growth rate $d_* = 4/(L^2r(2-r))$; the relative correction begins $1 + \beta_1/N + \beta_2/N^2 + \beta_3/N^3$ with the values of Section 17 (the draft's index n is N here). The decimals evaluate finite exact prescriptions and are not fitted. The leading asymptotic already in the entry, Kotěšovec's (18 February 2017, updated 20 April 2024), is not to be presented as new, nor described as labelled a conjecture.

A Provenance and editorial notes

[Added 3 October 2026, batch 85B.]

Sources. The report was built from two manuscripts of batch 85 of ProveIt's incoming-reports intake, written independently on the same subject.

- *Manuscript 06* (Part I, the base): the archive `Matrix_Compositions_Uniform_Asymptotics.zip` (21 files), main file `matrix_compositions.tex` (2,078 lines, 100 labels), a 31-page PDF titled *Matrix Compositions: Exact Spectra, Hankel Products, and Uniform Asymptotics*, dated October 3, 2026 (Pacific time). Its author line and PDF metadata read “OpenAI ChatGPT”, with “Research prepared for Vladimir Reshetnikov”. Its pin is the search-index commit `ce37e13f4aa16819c87c3ccc611362d15758f2ac` (2026-10-03, 17:05) [7]. It arrived in commit `345284ac8`.
- *Manuscript 01* (Part II): the archive `OEIS_Matrix_Compositions_Research.zip` (26 files), main file `matrix_compositions.tex` (1,561 lines, 98 labels), a 22-page PDF titled *Matrix compositions: finite spectra, uniform asymptotics, and two Poisson limits*, dated 3 October 2026, author line “Research report prepared for Vladimir Reshetnikov”. Its pin is the tree revision `d68b65ea064084fb121df9bb431b21d086f7be81`; it also records the search-index commit `ce37e13f4` [9]. It arrived in commit `317c1ce2e`, with the other archives of batch 85; the placement of batch 85A (`713149ded`) held it back so that it could be merged with manuscript 06.

Both were placed together by commit `fa2f3e419`, which deleted both archives from `docs/incoming`; they survive in their arrival commits. Neither manuscript is an edition of the other: they share about 0.7% and 0.5% of their word 8-grams (bibliography entries and one phrase), and their pins, code and notation differ.

Why manuscript 06 is the base. On every theorem the two share, manuscript 06 has the weaker hypotheses or the more general statement: recurrence minimality for every column weight $u > 0$ (manuscript 01: $u = 1$); a pole bound uniform in k , also weighted (manuscript 01: a factor $k - 1$); the coverage expansion of the whole generating function in complex w (manuscript 01: $\mathbb{P}(M = 0)$ only); and the correct attribution of the recurrence's existence to [4, Proposition 25]. It also has more new mathematics (the Hankel determinant product, the sharp total-variation rate, the joint Gaussian–Poisson law, the column central limit theorem, the prescribed-column threshold). Manuscript 01 is more general only in the kernel lemma, and that lemma only serves compact rays, where both theorems coincide.

Where the merge had to choose.

- Shared results are printed once, in manuscript 06's form, with a bracketed note naming manuscript 01's version (Table 5). Manuscript 01's proofs that are the same argument are summarized in those notes, not reprinted; its genuinely different devices are recorded there (Bonferroni truncation for the coverage probability, the tail bound $|\mathbf{W}_n(w)| \leq e^{|w|}$, the direct uniqueness argument for the maximum on the circle) or printed in Part II where a new result needs them.
- Manuscript 01's k -dependent pole bound is printed as Remark 4.2, because it is sharper for $k \leq 4$.
- Manuscript 01's kernel theorem is printed in full in Part II as the general form of Lemma 7.2, and Remark 7.3 says so where the lemma stands. Manuscript 06's lemma is kept unchanged.
- Manuscript 06's moments of D in Corollary 7.5 are kept, with a pointer to the stronger Theorem 16.2.
- Manuscript 01's closed form of β_1 is printed as a note after Corollary 7.4, being the same expression; its fifty-digit constants and β_3 are in Part II.
- Manuscript 01's overstated claim about the order conjecture is narrowed (Section 11.2), its draft OEIS text is printed with the correction (Section 22), and its example (2.8) is credited to [4, Remark 26].
- Manuscript 01's research question on joint column laws is re-scoped as partly answered by manuscript 06; manuscript 06's Research question 2 is re-scoped as answered for the event $M = 0$ by Theorem 13.1.
- Manuscript 01's inversion section is printed with a note that it is an instance of the transseries volume's apparatus; manuscript 01's source manifest (its Appendix C) is replaced by this appendix and the report README, and its rebuild instructions by the README.
- Manuscript 01's three tables (Tables 6, 7 and 8) are typeset, unchanged, from its shipped table bodies, the files `data/01-two-poisson-X.tex` with X one of `defect_table`, `error_table` and `coupon_table`. Its fourth body, `diagonal_table` (the same diagonal errors at more indices and digits), was not used by its article and is not used here.

What the write changed in manuscript 06's text.

- Every label gained the prefix `mxc`: (100 labels; every reference was updated); nothing was renamed otherwise and nothing was removed.
- Added: the second author and date lines of the title, the PDF metadata, the editorial note after the abstract, the part heading, Section 1.5 with Table 2, Remarks 2.3, 4.2 and 7.3, bracketed notes marked “[Added 3 October 2026, batch 85B]” in Sections 1, 2, 5, 7, 9 and 10, Part II, this appendix, and five bibliography entries [9, 10, 11, 12, 8].
- The figure path is now `figures/06-hankel-coupon_window.pdf`.
- No delivered sentence of manuscript 06 was changed or removed. Every non-claim is kept: the abstract's last sentence; the scope table's caption; the bounded audit; “The theorem controls the unrestricted count $A(n, k)$ uniformly” and what it does not say; the moving-mean remark; the compactness paragraph after Corollary 7.5; the prescribed-column caveat; the Hankel caveat on total positivity; the numerics caveats; and the caveat on the research questions.

Manuscript 01's text is restated in manuscript 06's letters (Table 2), with some proofs condensed; none of its statements was strengthened, and every one of its non-claims is kept (Section 11, the remarks after Theorems 13.1 and 16.2, Remark 15.1, Sections 18, 19, 21 and 22).

Overloaded letters in manuscript 06. Manuscript 06 reuses some letters; none is renamed, and in each case the section determines the meaning. D is $\binom{k+1}{2}$ in Section 2, the constant $D_u = 16b/L_u^2$ in Section 4, the function $D(\eta)$ in Section 5 and $n - Q$ in Section 7.6. d is $\binom{k+1}{2}$ and c is $\binom{k+1}{3}$ in Section 3, beside the growth constants d_k, c_k . b is $u/(1+u)$ in Sections 2–4, also u in Section 3, the variance κ_2 in Section 7, and b_j are factorial moments. $P_j(t)$ in Section 3 and $P_j(w)$ in Section 7.4 are different polynomials; $E_n(w)$ is a product, $E_j(\lambda)$ a saddle coefficient and $E_n = B\rho^{n-1}$ an error budget; r is $r_k = 1 - a_k$, the saddle, and a summation index; v is $Lr/2$ in Section 7 but a tilting parameter in the proof of Theorem 6.2; and m indexes d_m and counts columns.

Checks by the intake. The placement reran both packages on copies: manuscript 06's suite passed with its outputs byte-identical or equal apart from line endings and timings and its figure pixel-identical; manuscript 01's suite passed likewise, and its checksum ledger matched 25/25. Manuscript 06's diagonal constants and manuscript 01's agree in every printed digit, and their coverage probabilities at $k = 20, n = 60$ agree. The intake also verified Proposition 25 and Remark 26 in the published paper, the impulse of Remark 2.4 at $k = 1, 2$, the instances $H_{3,s}(u)$, $H_{2,1}(1) = 8$ and $H_{3,1}(1) = 62208$, the constants of Theorem 4.1 and Remark 4.2, the moments in Corollary 7.5, the increment expansion (2.11), the cumulant recursion at $j = 3$, and P_3 numerically. These checks do not replace an independent review of the proofs.

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